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M3ddisp functions

Synopsis

The functions prefixed with M3ddisp make up the 3D Display module. The 3D Display module allows you to display both point clouds and 3D objects created using the 3dgra module.

Functions

M3ddispAlloc
M3ddispAllocResult
M3ddispControl
M3ddispCopy
M3ddispCopyResult
M3ddispFree
M3ddispGetHookInfo
M3ddispGetResult
M3ddispGetView
M3ddispHookFunction
M3ddispInquire
M3ddispLut
M3ddispPick
M3ddispPickRect
M3ddispSelect
M3ddispSelectWindow
M3ddispSetHookInfo
M3ddispSetView

M3ddispAlloc

Synopsis

Allocate a 3D display or picking context.

Syntax

```
AIL_ID M3ddispAlloc(
    AIL_ID SysId,                  //in
    AIL_INT64 DispNum,             //in
    AIL_CONST_TEXT_PTR DispFormat, //in
    AIL_INT64 InitFlag,            //in
    AIL_ID *Disp3dIdPtr            //out
)
```

Description

This function allocates a 3D display or picking context on the specified system so that it can be used by subsequent Aurora Imaging Library3D display functions.

A 3D display allows you to display a point cloud, depth map, and/or 3D graphics list. To do so, use M3ddispSelect(). You can manipulate the view of a 3D display, either interactively using the keyboard and mouse or within your application using M3ddispSetView() and M3ddispControl(). You can change the keyboard control bindings, or disable the ability to manipulate the view interactively, using M3ddispControl() with M_ACTION_KEY_... or M_KEYBOARD_USE / M_MOUSE_USE, respectively. You can also control other aspects of the 3D display using M3ddispControl().

If you are developing an Aurora Imaging Library application that also acts as an Aurora Imaging Web Library server, you can allocate a display that can be presented in one or more instances of an Aurora Imaging Web Library client application (for example, running in a compatible web browser). To do so, use M_WEB when allocating the display. This type of display is referred to as an Aurora Imaging Web Library display. See the Fundamentals for creating your Aurora Imaging Web Library client application section of Chapter 58: Using Aurora Imaging Web Library to monitor your application for more information.

After allocating the 3D display or picking context, you should check if the operation was successful, using MappGetError() or by verifying that the 3D display or picking context identifier returned is not **M_NULL** (or *nullptr* if M_UNIQUE_ID was specified).

When the 3D display or picking context is no longer required, release it using M3ddispFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 3D display or picking context's lifetime and you must not manually free it.

Parameters

SysId

Specifies the system on which to allocate the 3D display or picking context. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

DispNum

Reserved for future expansion and must be set to **M_DEFAULT**.

DispFormat

Reserved for future expansion and must be set to AIL_TEXT(" **M_DEFAULT** ").

InitFlag INQ

Specifies the type of 3D display or picking context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the display type or picking context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_WINDOWED.	c		e	f	g	a		i		j	k	l	m	n	b
M_EXCLUSIVE	Specifies to present the 3D display full-screen, without a windowed border, in one of the Windows desktop screens. To use an exclusive display, your Windows desktop should be using more than one screen. Allocating an exclusive display on the main screen might not be convenient; if you do so, the Aurora Imaging Library exclusive display will hide third-party applications designed to start on the main screen and hide the Windows taskbar.	c		e	f	g	a		i		j	k	l	m	n	b
M_PICKING_AREA_CONTEXT	Specifies to allocate a picking area context that can be used with M3ddispPickRect() to pick a 3D graphic in a given rectangular region in a 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_PICKING_CONTEXT	Specifies to allocate a picking context that can be used with M3ddispPick() to pick a 3D graphic at a given position in a 3D display.	c		e	f	g	a		i		j	k	l	m	n	b

M_WEB	Specifies to allocate a 3D display that is presented upon selection in one or more instances of an Aurora Imaging Web Library client application (for example, running in a compatible web browser). To use an Aurora Imaging Web Library display, you must enable Aurora Imaging Web Library server functionality in your application using MappControl() with M_WEB_CONNECTION. You must also publish the display using MobjControl() with M_WEB_PUBLISH. An Aurora Imaging Web Library display cannot be presented by your Aurora Imaging Library application. It can only be presented by a connected instance of a client application (running on the local computer or a remote computer). In the client application, the 3D display is treated as thought it were a 2D display (for example, it has the object type `AIWL.M_DISPLAY` and you control it using AIWL.MdispControl or AIWL::MdispControl) See the Fundamentals for creating your Aurora Imaging Web Library client application section of Chapter 58: Using Aurora Imaging Web Library to monitor your application for more information.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOWED	Specifies to allocate a 3D display that is presented upon selection in its own window on the Windows desktop screen(s).	c		e	f	g	a		i		j	k	l	m	n	b

Disp3dIdPtr

Specifies the address of the variable in which to write the 3D display or picking context identifier or specifies the data type that the function should use to return the 3D display or picking context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D display or picking context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D display or picking context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DDISP_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D display or picking context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the 3D display identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated 3D display. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the picking area context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated picking area context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the picking context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated picking context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b	

Return value

The returned value is the 3D display or picking context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DDISP_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispAllocResult

Synopsis

Allocate a picking result buffer.

Syntax

```
AIL_ID M3ddispAllocResult(
    AIL_ID SysId,             //in
    AIL_INT64 ResultType,     //in
    AIL_INT64 ControlFlag,    //in
    AIL_ID *Result3ddispIdPtr //out
)
```

Description

This function allocates a picking result buffer on the specified system, to store results from an M3ddispPick() or M3ddispPickRect() operation.

After allocating the picking result buffer, you should check if the operation was successful, using MappGetError() or by verifying that the picking result buffer identifier returned is not **M_NULL** (or *nullptr* if M_UNIQUE_ID was specified).

When the picking result buffer is no longer required, release it using M3ddispFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the picking result buffer's lifetime and you must not manually free it.

Parameters

SysId

Specifies the system on which to allocate the picking result buffer. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Libraryapplication.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result buffer to allocate.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of result buffer

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PICKING_AREA_RESULT	Specifies to allocate a picking area result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_PICKING_RESULT	Specifies to allocate a picking result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Result3ddispIdPtr

Specifies the address of the variable in which to write the picking result buffer identifier or specifies the data type that the function should use to return the picking result buffer identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated picking result buffer ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated picking result buffer ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DDISP_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the picking result buffer (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the picking area result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated picking area result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the picking result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated picking result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the picking result buffer identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DDISP_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispControl

Synopsis

Control a 3D display or picking context setting.

Syntax

```
void M3ddispControl(
    AIL_ID Disp3dId,        //out
    AIL_INT64 ControlType,  //in
    AIL_DOUBLE ControlValue //in
)
```

Description

This function allows you to control the specified Aurora Imaging Library 3D display or picking context setting.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **M3ddispControl()** internally calls the *AIL_DOUBLE* version of this function (M3ddispControlDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (M3ddispControlInt64()).

Parameters

Disp3dId

Specifies the identifier of the target 3D display or picking context, previously allocated using M3ddispAlloc().

ControlType

Specifies the type of 3D display or picking context setting to control.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the new value to assign to the 3D display or picking context setting specified by the ControlType parameter.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- For controlling keyboard and mouse controls
- For general 3D display settings
- For controlling a picking context

The following ControlType and corresponding ControlValue parameter settings are used to enable, disable, or change keyboard and mouse bindings used to control of the 3D display. The default keyboard and mouse bindings are listed under M_KEYBOARD_USE and M_MOUSE_USE.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling keyboard and mouse controls

ControlType / ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ACTION_KEY_AUTO_ROTATE	Sets the keyboard keys that you can press to start and stop auto-rotation around the auto-rotation axis (set using M3dgraCopy() with the identifier of a 3D line geometry object). This action is equivalent to enabling or disabling auto-rotation using M_AUTO_ROTATE. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ															
M_DEFAULT	Same as M_KEY_R.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_R +	Specifies the R key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORBIT_DOWN		Sets the keyboard keys that you can press to orbit the viewpoint down around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL and a negative value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_DOWN.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORBIT_LEFT		Sets the keyboard keys that you can press to orbit the viewpoint left around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL and a negative value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_LEFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORBIT_RIGHT		Sets the keyboard keys that you can press to orbit the viewpoint right around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL and a positive value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_RIGHT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORBIT_UP		Sets the keyboard keys that you can press to orbit the viewpoint up around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL and a positive value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_UP.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ORIENTATION_BOTTOM_TILTED		Sets the keyboard keys that you can press to set the orientation of the viewpoint so that the view faces the bottom-front of the interest point. This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_BOTTOM_TILTED. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_1 +	Specifies the 1 key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_BOTTOM_VIEW		Sets the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly below the interest point (parallel to the Z-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_BOTTOM_VIEW. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_2.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_2 +	Specifies the 2 key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_FRONT_VIEW		Sets the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly in front of the interest point (parallel to the Y-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_FRONT_VIEW. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_5.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_5 +	Specifies the 5 key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ORIENTATION_LEFT_VIEW		Sets the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly to the left of the interest point (parallel to the X-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_LEFT_VIEW. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_4.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_4 +	Specifies the 4 key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_REAR_VIEW		Sets the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly behind the interest point (parallel to the Y-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_REAR_VIEW. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_3.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_3 +	Specifies the 3 key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_RIGHT_VIEW		Sets the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly to the right of the interest point (parallel to the X-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_RIGHT_VIEW. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_6.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_6 +	Specifies the 6 key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ORIENTATION_TOP_TILTED		Sets the keyboard keys that you can press to set orientation of the viewpoint so that the view faces the top-front of the interest point. This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_TOP_TILTED. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_7.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_7 +	Specifies the 7 key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_TOP_VIEW		Sets the keyboard keys that you can press to set orientation of the viewpoint so that it is directly above the interest point (parallel to the Z-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_TOP_VIEW. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_8.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_8 +	Specifies the 8 key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESET		Sets the keyboard keys that you can press to reset the view to the current default position and orientation. This action is equivalent to using M3ddispSetView() with M_AUTO and M_TOP_TILTED. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_HOME.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ROLL_LEFT		Sets the keyboard keys that you can press to roll the view left. This action is equivalent to using M3ddispSetView() with M_ROLL + M_COMPOSE_WITH_CURRENT and a negative value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_LEFT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT + M_KEY_ALT	Specifies the Left Arrow key with the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ROLL_RIGHT		Sets the keyboard keys that you can press to roll the view right. This action is equivalent to using M3ddispSetView() with M_ROLL + M_COMPOSE_WITH_CURRENT and a positive value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_RIGHT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT + M_KEY_ALT	Specifies the Right Arrow key with the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_BACKWARD		Sets the keyboard keys that you can press to move the view backward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_DOWN + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN + M_KEY_ALT	Specifies the Down Arrow key with the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_DOWN		Sets the keyboard keys that you can press to move the view downward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_DOWN + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN + M_KEY_SHIFT	Specifies the Down Arrow key with the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_FORWARD		Sets the keyboard keys that you can press to move the view forward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_UP + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP + M_KEY_ALT	Specifies the Up Arrow key with the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_TRANSLATE_LEFT		Sets the keyboard keys that you can press to move the view left relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_LEFT + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT + M_KEY_SHIFT	Specifies the Left Arrow key with the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_RIGHT		Sets the keyboard keys that you can press to move the view right relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_RIGHT + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT + M_KEY_SHIFT	Specifies the Right Arrow key with the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_UP		Sets the keyboard keys that you can press to move the view upward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_UP + M_KEY_SHIFT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP + M_KEY_SHIFT	Specifies the Up Arrow key with the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TURN_DOWN		Sets the keyboard keys that you can press to pan the view downward by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL + M_MOVE_INTEREST_POINT and a positive value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_KEY_ARROW_DOWN + M_KEY_SHIFT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN + M_KEY_SHIFT + M_KEY_ALT	Specifies the Down Arrow key with the Shift key and Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_TURN_LEFT		Sets the keyboard keys that you can press to pan the view left by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL + M_MOVE_INTEREST_POINT and a positive value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_LEFT + M_KEY_SHIFT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT + M_KEY_SHIFT + M_KEY_ALT	Specifies the Left Arrow key with the Shift key and Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TURN_RIGHT		Sets the keyboard keys that you can press to pan the view right by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL + M_MOVE_INTEREST_POINT and a negative value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_RIGHT + M_KEY_SHIFT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT + M_KEY_SHIFT + M_KEY_ALT	Specifies the Right Arrow key with the Shift key and Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TURN_UP		Sets the keyboard keys that you can press to pan the view upward by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL + M_MOVE_INTEREST_POINT and a negative value. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ARROW_UP + M_KEY_SHIFT + M_KEY_ALT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP + M_KEY_SHIFT + M_KEY_ALT	Specifies the Up Arrow key with the Shift key and Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ZOOM_IN		Sets the keyboard keys that you can press to move the viewpoint closer to the interest point. This action is equivalent to using M3ddispSetView() with M_DISTANCE and a value greater than 1.0. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_ADD.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ADD +	Specifies the Add key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ZOOM_OUT		Sets the keyboard keys that you can press to move the viewpoint further from the interest point. This action is equivalent to using M3ddispSetView() with M_DISTANCE and a value between 0.0 and 1.0. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_SUBTRACT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b

	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_MODIFIER_SPEED		Sets the keyboard key that you can use to modify the movement speed for all actions when using other keyboard commands. You can set the factor by which the speed will be modified using M_ALTERNATE_SPEED_FACTOR. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_CTRL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ALT +	Specifies the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CTRL +	Specifies the CTRL key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SHIFT +	Specifies the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALTERNATE_SPEED_FACTOR		Sets by which factor to multiply the movement speed when the key set using M_ACTION_MODIFIER_SPEED is held. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 3.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the factor by which to multiply the movement speed.	c		e	f	g	a		i		j	k	l	m	n	b

M_FUNCTION_KEY_CYCLE_APPEARANCES		Sets the keyboard keys that you can press to cycle through the given appearance modes for all point clouds currently being shown in the internal 3D graphics list of the 3D display. Note, for the appearance change to be visible, the point cloud must have a mesh component. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_CTRL + M_KEY_M.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CTRL + M_KEY_M +	Specifies the Ctrl key with the M key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_FUNCTION_KEY_POINTSIZE_DECREMENT		Sets the keyboard keys that you can press to decrement the point size (M_THICKNESS) for all point clouds currently being displayed. This only affects point clouds which have M_APPEARANCE set to M_POINTS, or which do not have a mesh. For each point cloud, M_THICKNESS is decremented independently based on the current value for that point cloud. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_Z +	Specifies the Z key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_FUNCTION_KEY_POINTSIZE_INCREMENT		Sets the keyboard keys that you can press to increment the point size (M_THICKNESS) for all point clouds currently being displayed. This only affects point clouds which have M_APPEARANCE set to M_POINTS, or which do not have a mesh. For each point cloud, M_THICKNESS is incremented independently based on the current value for that point cloud. If the keyboard key that you specify is used by another action setting (M_ACTION_KEY_...), Aurora Imaging Library sets that other action setting to M_NONE. This setting only has an effect if you enable M_KEYBOARD_USE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_KEY_X.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_X +	Specifies the X key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_FUNCTION_KEY_TOGGLE_LOD		Sets the keyboard keys that you can press to cycle through the given level of detail (LoD) modes for all point clouds currently being shown in the internal 3D graphics list of the 3D display. The LoD mode will cycle through the three modes (M_DISABLE, M_ENABLE_ON_ACTION, and M_ENABLE_ALWAYS). INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_KEY_L.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_L +	Specifies the L key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEYBOARD_USE		Sets whether the user can interactively rotate, orbit, and scroll the view or change the point size and appearance mode for the 3D display using the keyboard. The default key usage is: INQ	c		e	f	g	a		i		j	k	l	m	n	b

Up arrow	Orbits the viewpoint up around the interest point (relative to the roll of the view).
Down arrow	Orbits the viewpoint down around the interest point (relative to the roll of the view).
Left arrow	Orbits the viewpoint left around the interest point (relative to the roll of the view).
Right arrow	Orbits the viewpoint right around the interest point (relative to the roll of the view).
Alt-Up arrow	Moves the view forward by moving both the viewpoint and interest point.
Alt-Down arrow	Moves the view backward by moving both the viewpoint and interest point.
Alt-Left arrow	Rolls the view left.
Alt-Right arrow	Rolls the view right.
Shift-Up arrow	Scrolls the view up by moving the viewpoint and interest point.

Shift-Down Arrow	Scrolls the view down by moving the viewpoint and interest point.
Shift-Left arrow	Scrolls the view left by moving the viewpoint and interest point.
Shift-Right arrow	Scrolls the view right by moving the viewpoint and interest point.
Alt-Shift-Up arrow	Rotates the view up by moving the interest point.
Alt-Shift-Down arrow	Rotates the view down by moving the interest point.
Alt-Shift-Left arrow	Rotates the view left by moving the interest point.
Alt-Shift-Right arrow	Rotates the view right by moving the interest point.
+	Moves the viewpoint towards the interest point (zoom in).
-	Moves the viewpoint away from the interest point (zoom out).
Home	Resets the view to the default position and orientation.

		r	Enables or disables auto-rotation.													
		Z	Decrement the point size in the display.													
		X	Increment the point size in the display.													
	M_DEFAULT	Same as M_ENABLE.		c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the keyboard to control the view is disabled.		c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the keyboard to control the view is enabled.		c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_ROLL		Sets whether the user can interactively roll the view for the 3D display using the mouse. INQ		c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.		c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to roll the view is not enabled.		c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to roll the view is enabled.		c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_ROTATION		Sets whether the user can interactively rotate the view for the 3D display using the mouse. INQ		c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.		c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to rotate the view is not enabled.		c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that using the mouse to rotate the view is enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_TRANSLATION		Sets whether the user can interactively translate the view for the 3D display using the mouse. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to translate the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to translate the view is enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_TRANSLATION_X		Sets whether the user can interactively translate the view for the 3D display in the X-direction using the mouse. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to translate the view in the X-direction is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to translate the view in the X-direction is enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_TRANSLATION_Y		Sets whether the user can interactively translate the view for the 3D display in the Y-direction using the mouse. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to translate the view in the Y-direction is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to translate the view in the Y-direction is enabled.	c		e	f	g	a		i		j	k	l	m	n	b

M_MOUSE_USE	Sets whether the user can interactively rotate, orbit, and scroll the view for the 3D display using the mouse. The default mouse usage is: INQ Left mouse button + move the mouse: Orbits the viewpoint around the interest point. Right mouse button + move the mouse: Scrolls the view by moving the viewpoint and interest point. Middle mouse button + move the mouse: Rolls the view. Scroll wheel up: Moves the viewpoint towards the interest point. Scroll wheel down: Moves the viewpoint away from interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that using the mouse to control the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that using the mouse to control the view is enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_ZOOM	Sets whether the user can interactively zoom the 3D display using the mouse. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to zoom the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to zoom the view is enabled.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are available for 3D displays.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For general 3D display settings																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	
M_ASSOCIATED_GRAPHIC_LIST_ID		Sets the 2D graphics list to associate with the 3D display. When a list is associated it must have no graphics set in world units and it cannot have any graphics with calibration set. INQ																
	M_DEFAULT	Same as M_NULL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Specifies that no 2D graphics list is associated with the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
	2D graphics list identifier	Specifies the identifier of the 2D graphics list to associate with the 3D display. The 2D graphics list must have been previously allocated using MgraAllocList().	c		e	f	g	a		i		j	k	l	m	n	b

M_AUTO_ROTATE	Sets whether to automatically rotate the view (both the viewpoint and interest point) around the auto-rotation axis. This gives the impression that the scene is rotating. By default, the auto-rotation axis is that which is parallel to the Z-axis and intersects the default position of the interest point (typically the center of the first 3D graphic added to the internal 3D graphics list of the 3D display). You can change the auto-rotation axis using M3ddispCopy() with M_ROTATION_AXIS and the identifier of a line 3D geometry object. Typically, you should only use an auto-rotation axis that intersects the interest point. Some users might be disoriented by the view rotating around a different axis, especially when the rotation indicator is enabled (using M_ROTATION_INDICATOR). Instead of this control type, you can also use M_ACTION_KEY_AUTO_ROTATE to enable/disable rotation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies to not auto-rotate the view.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to auto-rotate the view.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_COLOR	Sets the primary background color for the 3D display, used when M_BACKGROUND_MODE is set to M_SINGLE_COLOR or M_GRADIENT_VERTICAL. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies a dark gray. This is equivalent to M_RGB888() with the values (0,35,40).	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to which the background color will be set.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters														
	Red Specifies the red band, as a value between 0 and 255.														
	Green Specifies the green band, as a value between 0 and 255.														

		Blue Specifies the blue band, as a value between 0 and 255.															
	M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_COLOR_GRADIENT		Sets the secondary background color for the 3D display, used when M_BACKGROUND_MODE specifies that the background is a gradient. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies a light gray. This is equivalent to M_RGB888() with the values (90,110,130).	c		e	f	g	a		i		j	k	l	m	n	b
	M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to which the background color will be set.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Red Specifies the red band, as a value between 0 and 255.															
		Green Specifies the green band, as a value between 0 and 255.															
		Blue Specifies the blue band, as a value between 0 and 255.															
	M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_MODE		Sets whether the background of the 3D display is a solid color or a gradient. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_GRADIENT_VERTICAL.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BACKGROUND_IMAGE	Specifies that the background is an image, set using M3ddispCopy() with M_BACKGROUND_IMAGE. If no background image is set, the background is shown as though this value were set to M_GRADIENT_VERTICAL.	c		e	f	g	a		i		j	k	l	m	n	b

	M_GRADIENT_VERTICAL	Specifies that the background is a gradient. The top of the window is the color specified by M_BACKGROUND_COLOR and the bottom of the window is the color specified by M_BACKGROUND_COLOR_GRADIENT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SINGLE_COLOR	Specifies that the background is a solid color, specified by M_BACKGROUND_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_FOV_HORIZONTAL_ANGLE		Sets the horizontal field of view of the 3D display as an angle. This is the angle that defines how much of the scene is shown within the width of the 3D display. The default value is whichever value will give the field of view the same aspect ratio as the 3D display's window size, given the default vertical field of view. Note that the display's M_FOV_HORIZONTAL_ANGLE and M_FOV_VERTICAL_ANGLE must always have the same aspect ratio as M_SIZE_X and M_SIZE_Y. Any time one of these values is changed, M_FOV_HORIZONTAL_ANGLE and/or M_FOV_VERTICAL_ANGLE will be automatically updated to maintain this coherence. For example, if you increase M_SIZE_Y, M_FOV_VERTICAL_ANGLE will also be increased by an appropriate amount to maintain the aspect ratio. It is possible for either FoV angle to exceed 180 degrees, either because the window was resized or because the shorter window axis is set to a high FoV angle. If this happens, the FoV angles are clamped such that the ratio is maintained, but neither FoV angle exceeds 179.999 degrees. Once the window size is reduced, the actual FoV angle that was requested will again be used. The "temporary" angle will be used when returning the FoV angle for either axis; if the vertical angle is clamped, inquiring either the vertical or horizontal FoV angle will return values which reflect what is currently shown on-screen, not the values that you would get without the clamping. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* < 180.0	Specifies the horizontal field of view, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_FOV_VERTICAL_ANGLE	Sets the vertical field of view of the 3D display as an angle. This is the angle that defines how much of the scene is shown within the height of the 3D display. Note that the display's M_FOV_HORIZONTAL_ANGLE and M_FOV_VERTICAL_ANGLE must always have the same aspect ratio as M_SIZE_X and M_SIZE_Y. Any time one of these values is changed, M_FOV_HORIZONTAL_ANGLE and/or M_FOV_VERTICAL_ANGLE will be automatically updated to maintain this coherence. For example, if you increase M_SIZE_Y, M_FOV_VERTICAL_ANGLE will also be increased by an appropriate amount to maintain the aspect ratio. It is possible for either FoV angle to exceed 180 degrees, either because the window was resized or because the shorter window axis is set to a high FoV angle. If this happens, the FoV angles are clamped such that the ratio is maintained, but neither FoV angle exceeds 179.999 degrees. Once the window size is reduced, the actual FoV angle that was requested will again be used. The "temporary" angle will be used when returning the FoV angle for either axis; if the vertical angle is clamped, inquiring either the vertical or horizontal FoV angle will return values which reflect what is currently shown on-screen, not the values that you would get without the clamping. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 30°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* < 180.0	Specifies the vertical field of view, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_AVERAGE_SMOOTHING	Sets how many frames to use when calculating the average frametime for M3ddispInquire() with M_FRAMETIME_AVERAGE or M3ddispHookFunction() with M_FRAMETIME_HIGH or M_FRAMETIME_LOW. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 3.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of frames.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_THRESHOLD_HIGH	Sets the frametime above which to generate an M_FRAMETIME_HIGH event. If you try to set this lower than the current value of M_FRAMETIME_THRESHOLD_LOW, it will instead be set to the same value. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 0.066.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the frametime, in seconds.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_THRESHOLD_LOW		Sets the frametime below which to generate an M_FRAMETIME_LOW event. If you try to set this higher than the current value of M_FRAMETIME_THRESHOLD_HIGH, it will instead be set to the same value. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.016.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the frametime, in seconds.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_OPACITY		Sets the opacity level of annotations generated from the 2D graphics list associated with the 3D display. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to use partial opacity. This is equivalent to setting an opacity of 100.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* <= 100	Specifies the opacity (as a percentage), where 0 is completely transparent and 100 is completely opaque.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_INPUT_MODE		Sets how the 3D display will handle user inputs. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NORMAL_3D.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ASSOCIATED_GRAPHIC_LIST_2D	Specifies to use the mouse movements and keystrokes to draw graphics into the associated 2D graphics list (M_ASSOCIATED_GRAPHIC_LIST_ID).	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMAL_3D	Specifies to use the mouse movements and keystrokes to manipulate the view of the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b

M_LOD_DEGRADATION_LEVEL		Sets the point/pixel density at which the display will choose a lower level of detail (LoD) for a given point cloud. A lower value will reduce the LoD more aggressively, improving performance at the cost of visible quality reduction. For example, a value below 1 will show lower LoDs even when the view is zoomed in, and values over 4 will rarely show lower LoDs. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/pixel density.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOD_DEGRADE_MODE		Sets which level of detail (LoD) degradation mode to use. Note, this setting only applies when M_VIEW_BASED_LOD is set to M_ENABLE INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the LoD change.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Same as M_ENABLE_ON_ACTION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE_ALWAYS	Specifies to always show a degraded LoD.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE_ON_ACTION	Specifies to use one LoD lower when the user is interacting with the display. Manipulating the view with the mouse, keyboard, or an Aurora Imaging Library function, manipulating a graphics handle, and automatic rotations are all considered interactions with the display. The display will wait for at least as many seconds specified by M_LOD_TIME_TO_SHOW_DEGRADED before rendering a full detail view.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOD_TIME_TO_SHOW_DEGRADED		Sets how long to wait after an automatically degraded render, before rendering at the normally selected LoD. Note, this has no effect if M_LOD_DEGRADE_MODE is set to M_DISABLE INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies how long to wait, in seconds.	c		e	f	g	a		i		j	k	l	m	n	b
M_PARALLEL_SCALE		Sets the parallel scale when using the M_PARALLEL_PRECISE projection mode. The parallel scale is the ratio between world units and pixels. For example, if the parallel scale is 2, a plane with a size of 200x200 units will appear as a square of 100x100 pixels (assuming it is viewed directly from the top). As parallel scale increases, the display will zoom out. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the parallel scale.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROJECTION_MODE		Sets the mode of projection, which affects the appearance of 3D graphics shown in the 3D display. The perspective projection mode accounts for perspective to provide information about depth; this is similar to how humans see 3D objects. Parallel projection modes ignore perspective to allow for more accurate measurements. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_PERSPECTIVE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PARALLEL	Specifies to use parallel projection. In this mode, parallel lines remain parallel and the 3D display gives a "blueprint" view. This is useful when trying to determine how two points are aligned with one another, particularly when used with a view that is aligned with the object (for example, top-down or side-on). Unlike M_PARALLEL_PRECISE, you do not specify a parallel scale. The parallel scale is automatically calculated so that the plane defined by the view direction and interest point has approximately the same size in parallel projection as it would in perspective projection, given the current window size and M_FOV_HORIZONTAL_ANGLE / M_FOV_VERTICAL_ANGLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_PARALLEL_PRECISE	Specifies to use parallel projection, with a defined ratio between world units and pixel size. In this mode, parallel lines remain parallel and the 3D display gives a "blueprint" view. This is useful when trying to determine how two points are aligned with one another, particularly when used with a view that is aligned with the object (for example, top-down or side-on). You can specify the ratio between world units and pixels using M_PARALLEL_SCALE. Note that when this mode is specified, the keyboard and mouse wheel control the parallel scale instead of the view distance. The distance of a point from the viewpoint is irrelevant; it will always be shown at the same position in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PERSPECTIVE	Specifies to use perspective projection. In this mode, the 3D display has perspective distortion. Parallel lines appear to converge and points that are further away appear smaller and closer to the center of the view. Perspective projection does not preserve relative proportions, but it works in a similar way to the human visual system and provides a more natural view than parallel projection. When using this mode, you can specify an angle to define what is visible (M_FOV_..._ANGLE). This is analogous to the focal length of a camera.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATION_INDICATOR		Sets whether to show the rotation indicator in the center of the view. The rotation indicator is always positioned at the interest point, and axis-aligned with the working coordinate system of the 3D display. This is useful for showing the current orientation of the view. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE_ON_MOUSE_CLICK.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to not show the rotation indicator.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to always show the rotation indicator.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE_ON_MOUSE_CLICK	Specifies to show the rotation indicator when the left mouse button is held down.	c		e	f	g	a		i		j	k	l	m	n	b

M_ROTATION_SPEED	Sets how fast, and in what direction, the view rotates when M_AUTO_ROTATE is set to M_ENABLE. This is the number of degrees the view rotates per second. If this value is set to a positive number, the view will auto-rotate to the right. Otherwise, it will auto-rotate to the left. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 20 degrees/sec.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the auto-rotation speed, in degrees/sec.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X	Sets the horizontal resolution of the window. Note that the display's M_FOV_HORIZONTAL_ANGLE and M_FOV_VERTICAL_ANGLE must always have the same aspect ratio as M_SIZE_X and M_SIZE_Y. Any time one of these values is changed, M_FOV_HORIZONTAL_ANGLE and/or M_FOV_VERTICAL_ANGLE will be automatically updated to maintain this coherence. For example, if you increase M_SIZE_Y, M_FOV_VERTICAL_ANGLE will also be increased by an appropriate amount to maintain the aspect ratio. It is possible for either FoV angle to exceed 180 degrees, either because the window was resized or because the shorter window axis is set to a high FoV angle. If this happens, the FoV angles are clamped such that the ratio is maintained, but neither FoV angle exceeds 179.999 degrees. Once the window size is reduced, the actual FoV angle that was requested will again be used. The "temporary" angle will be used when returning the FoV angle for either axis; if the vertical angle is clamped, inquiring either the vertical or horizontal FoV angle will return values which reflect what is currently shown on-screen, not the values that you would get without the clamping. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 800.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the horizontal resolution, in pixels. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_Y	Sets the vertical resolution of the window. Note that the display's M_FOV_HORIZONTAL_ANGLE and M_FOV_VERTICAL_ANGLE must always have the same aspect ratio as M_SIZE_X and M_SIZE_Y. Any time one of these values is changed, M_FOV_HORIZONTAL_ANGLE and/or M_FOV_VERTICAL_ANGLE will be automatically updated to maintain this coherence. For example, if you increase M_SIZE_Y, M_FOV_VERTICAL_ANGLE will also be increased by an appropriate amount to maintain the aspect ratio. It is possible for either FoV angle to exceed 180 degrees, either because the window was resized or because the shorter window axis is set to a high FoV angle. If this happens, the FoV angles are clamped such that the ratio is maintained, but neither FoV angle exceeds 179.999 degrees. Once the window size is reduced, the actual FoV angle that was requested will again be used. The "temporary" angle will be used when returning the FoV angle for either axis; if the vertical angle is clamped, inquiring either the vertical or horizontal FoV angle will return values which reflect what is currently shown on-screen, not the values that you would get without the clamping. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 600.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the vertical resolution, in pixels. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_SYNCHRONIZE	Synchronizes this thread with in an internal thread by forcing the thread to block until all previously requested operations on the display have been processed.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b
M_TITLE	Sets the display's title to the specified string. If no string is specified, the window is given a default title ("Aurora Imaging Library Display #x"). INQ	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*WindowTitle*") 1	Specifies the name of the window title (or display's title). > Note, if using a C compiler, you must pass the value as M_PTR_TO_AIL_INT(AIL_TEXT("*WindowTitle*")).	c		e	f	g	a		i		j	k	l	m	n	b

M_TRANSPARENCY_SORT_MODE	Sets which depth sorting method to use when semi-opaque (semi-transparent) 3D graphics are shown in the 3D display. More advanced modes can resolve visual errors where semi-opaque graphics are always shown in front of other graphics, regardless of which graphic is closer to the viewpoint. Note that this setting can alter the appearance of both opaque and semi-opaque co-planar surfaces (for example, if you create two plane graphics with the same position and normal). Graphics on higher layers will always be rendered completely in front of graphics on lower layers. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DEPTH_PEELING.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEPTH_PEELING	Specifies to use depth peeling. This mode resolves most transparency sorting errors, at the cost of increased GPU usage when there are multiple semi-opaque 3D graphics in the 3D graphics list of the 3D display. Typically, this mode does not affect performance if all graphics in the graphics list are fully opaque.	c		e	f	g	a		i		j	k	l	m	n	b
M_FAST	Specifies to use simple depth sorting. This mode offers maximum performance, but can exhibit significant visual errors when there are multiple semi-opaque 3D graphics in the 3D graphics list, particularly if they intersect with other graphics.	c		e	f	g	a		i		j	k	l	m	n	b
M_UPDATE	Sets whether Aurora Imaging Library will update the 3D display. This control type can be used to temporarily disable display updates when the internal 3D graphics list (or a buffer/container associated with a 3D graphic in the 3D graphics list) is being modified by more than one operation; the 3D display can then be updated at the end of all operations. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to update the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to update the 3D display. Also, the 3D display is forced to update immediately.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOW	Specifies to force an immediate update of the 3D display. In this case, the display is updated even if M_UPDATE is set to M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

M_UPDATE_GRAPHIC_LIST	Sets whether Aurora Imaging Library should update the 3D display when modifications are made to the 3D display's associated 2D graphics list. To associate a 2D graphics list to the 3D display, use M_ASSOCIATED_GRAPHIC_LIST_ID. With M_UPDATE_GRAPHIC_LIST, you can temporarily disable display updates (M_DISABLE) when the 2D graphics list is being modified by multiple operations; you can then enable display updates (M_ENABLE) at the end of all operations. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the 3D display is not updated when the associated 2D graphics list is modified.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the 3D display is immediately updated when the associated 2D graphics list is modified (refresh not required).	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_BOX_NODE	Sets on which 3D graphic the view will be focused when you use M3ddispSetView() with M_AUTO. If the specified 3D graphic is later removed from the 3D graphics list, the default setting is automatically restored. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0 (the label of the root node).	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the label of a 3D graphic in the 3D display's 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_INITIAL_POSITION_X	Sets the initial left-most X-coordinate of the window. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-coordinate, in pixels. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_INITIAL_POSITION_Y	Sets the initial top-most Y-coordinate of the window. Only integer values are accepted. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-coordinate, in pixels. Only integer values are accepted.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_MAXBUTTON		Sets whether the window's maximize button is visible. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the button is not visible.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the button is visible.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_MINBUTTON		Sets whether the window's minimize button is visible. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the button is not visible.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the button is visible.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_MOVE		Sets whether window movement is allowed. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to allow window movement.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to allow window movement. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_OVERLAP		Sets whether the window can be overlapped by another. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to allow the window to be overlapped by another (keep window on top).	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to allow the window to be overlapped by another. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

M_WINDOW_RESIZE	Sets whether window resizing is allowed. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to allow window resizing.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to allow window resizing. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_SHOW	Sets whether the window should be shown or hidden. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the window should be hidden.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the window should be shown. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_SYSBUTTON	Sets whether the window's system buttons are visible. The system buttons refer to the collection of the window's minimize, maximize, and close buttons. To individually set whether the minimize and maximize buttons are visible, change the M_WINDOW_MAXBUTTON or M_WINDOW_MINBUTTON control type settings. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the buttons are not visible.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the buttons are visible.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_TITLE_BAR	Sets whether the window's title bar is visible. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the title bar is not visible.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the title bar is visible. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

The following ControlType and corresponding ControlValue parameter settings are available for picking contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a picking context																	
ControlType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ControlValue																
M_BLOCKABLE	Sets whether 3D graphics are blocked from being picked if the closest 3D graphic at a given pixel is one that has been filtered out. For example, if you are only looking for boxes (M_GRAPHIC_TYPE set to M_GRAPHIC_TYPE_BOX), and there is a box and a line at a given pixel, but the line is in front of the box, enabling this control type will result in no detected 3D graphic, since the line blocks the box. Disabling this control type will cause the pick operation to ignore the line and detect the box. Note that enabling this control type might slow down the pick operation and negate the performance benefits of filtering out 3D graphics. When this control type is enabled, Aurora Imaging Library checks all 3D graphics (including those that have been filtered out) on a given layer to determine which ones were hit by the ray, and then identifies the one closest to the viewpoint. When this control type is disabled, Aurora Imaging Library will only check whether the ray hit any 3D graphics that have not been filtered out, resulting in a faster pick operation. Note, this setting is only available for point picking contexts. INQ															

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to ignore filtered out 3D graphics entirely. Aurora Imaging Library will not check whether the ray intersected with 3D graphics that have been filtered out.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to acknowledge filtered out 3D graphics. Aurora Imaging Library will check every 3D graphic to determine with which ones the ray intersected. The pick operation will not report a detected 3D graphic if the closest intersected 3D graphic on the highest layer has been filtered out.	c		e	f	g	a		i		j	k	l	m	n	b

M_FACES_AND_LINES	Sets whether faces and lines can be picked. Note that the faces of a 3D graphic with a wireframe appearance still exist and can be detected by the pick operation when this control is enabled. If you want to detect 3D graphics inside a wireframe box, you must ignore its faces by disabling this control (since its faces are still valid raycast targets, even though they are not visible). This control type only applies to meshed point cloud graphics, arc graphics, line graphics, lines graphics, filled graphics, and graphics with M_APPEARANCE set to M_SOLID or M_WIREFRAME. To control whether points can be picked, use M_POINTS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that faces and lines are not pickable.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that faces and lines are pickable.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE	Sets which type of 3D graphic can be detected by the pick operation. All other types of 3D graphics are filtered out and not pickable. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANY	Specifies any type of 3D graphic.	c		e	f	g	a		i		j	k	l	m	n	b

	M_GRAPHIC_TYPE_ARC	Specifies an arc graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_ARC_FILL	Specifies a filled arc graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_AXIS	Specifies an axis graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_BOX	Specifies a box graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_CYLINDER	Specifies a cylinder graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_DOTS	Specifies dots graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_GRID	Specifies a grid graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINE	Specifies a line graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINES	Specifies a lines graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_PLANE	Specifies a plane graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POINT_CLOUD	Specifies a point cloud graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POLYGON	Specifies a polygon graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_RECT	Specifies a rectangle graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_SPHERE	Specifies a sphere graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_TEXT	Specifies text graphic.	c		e	f	g	a		i		j	k	l	m	n	b

M_LABEL	Sets which label is required for a 3D graphic to be detected by the pick operation. All other 3D graphics are filtered out and not pickable. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_NO_LABEL.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_LABEL	Specifies to not filter out 3D graphics according to their label. The pick operation searches for all 3D graphics of the specified type, regardless of their label.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the label of the 3D graphic that is pickable.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY_THRESHOLD	Sets the minimum opacity required for a 3D graphic to be detected by the pick operation. 3D graphics with an opacity lower than the specified threshold are filtered out and are not pickable. Note that 3D graphics with an opacity of 0.0 and 3D graphics that are invisible are never pickable. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 100.0	Specifies the opacity threshold. 3D graphics with an opacity lower than the specified value are not pickable.	c		e	f	g	a		i		j	k	l	m	n	b

M_POINTS	Sets whether points can be picked. This control type only applies to point cloud graphics, dots graphics, and graphics with M_APPEARANCE set to M_POINTS. To control whether faces and lines can be picked, use M_FACES_AND_LINES. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that points are not pickable.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that points are pickable.	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER_MASK	Sets on which layers a 3D graphic can be detected by the pick operation. All 3D graphics on different layers are filtered out and not pickable. Graphics on higher layers are always detected before (and drawn in front of) those on lower layers. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
M_LAYER_MASK(AIL_INT64 Value)	Specifies to only detect 3D graphics on the specified layer.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

		Value Specifies the layer. The layer must be between 0 and 9. Note, you can combine multiple values using the bitwise operator, "|". For example, specify M_LAYER_MASK(0) | M_LAYER_MASK(4) | M_LAYER_MASK(5) to mask out all layers other than 0, 4, and 5.														
	M_LAYER_UNMASK(AIL_INT64 Value)	Specifies to detect 3D graphics on all but the specified layer.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters														
		Value Specifies the layer. The layer must be between 0 and 9. Note, you can combine multiple values using the bitwise operator, "&". For example, specify M_LAYER_UNMASK(0) & M_LAYER_UNMASK(4) & M_LAYER_UNMASK(5) to mask out layers 0, 4, and 5.														
	M_ANY	Specifies to detect 3D graphics on any layer.	c		e	f	g	a		i		j	k	l	m	n	b
	0*b*0000000000 <= *Value* <= 0*b*1111111111	Specifies a bit mask, where the status of bits from 0 to 9 indicate which layers to mask; only 3D graphics on the layers that correspond to bits with non-zero values will be detected. For example, you can set this value to 0b0010000010 (or (1 << 1) + (1 << 7)) to only detect 3D graphics on layers 1 and 7.	c		e	f	g	a		i		j	k	l	m	n	b

M_SELECTION_RADIUS		Sets the approximate radius around the specified position for the pick operation to search. Note, this setting is only available for point picking contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 8.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the selection radius, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the ControlValue parameter listed in For controlling keyboard and mouse controls. (exceptions might exist)

You can add one or more of the following values to the above-mentioned values to specify an additional keyboard key.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying additional keyboard keys																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_KEY_ALT	Specifies the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CTRL	Specifies the Ctrl key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_SHIFT	Specifies the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void M3ddispControlDouble (AIL_ID Disp3dId, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

Disp3dId

See Disp3dId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void M3ddispControlInt64 (AIL_ID Disp3dId, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

Disp3dId

See Disp3dId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispCopy

Synopsis

Copy data to or from a 3D display.

Syntax

```
void M3ddispCopy(
    AIL_ID SrcAilObjectId, //in
    AIL_ID DstAilObjectId, //out
    AIL_INT64 CopyType,    //in
    AIL_INT64 ControlFlag  //in
)
```

Description

This function copies data (such as a transformation matrix or background image) to or from a 3D display.

Parameters

SrcAilObjectId

Specifies the identifier of the source object. Note that if the source object is not a 3D display, the destination object must be.

See the Parameter associations section for possible values that can be specified.

DstAilObjectId

Specifies the identifier of the destination object. Note that if the destination object is not a 3D display, the source object must be.

See the Parameter associations section for possible values that can be returned.

CopyType

Specifies the type of copy operation to perform.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Specifies if the 3D display should be refreshed.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying if the display should be updated																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default behavior. If a view is modified a hook will be generated.	c		e	f	g	a		i		j	k	l	m	n	b

M_NO_HOOK	Prevents calling the M_VIEW_CHANGE hook after the function is completed. > This control flag is available for M_VIEW_MATRIX, with a 3D Display as the destination.	c		e	f	g	a		i		j	k	l	m	n	b

Parameter associations

The table below lists possible values for the **SrcAilObjectId** and **CopyType** parameters and possible values returned to the **DstAilObjectId** parameter.

- For specifying the copy type

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the copy type																	
CopyType			Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	SrcAilObjectId																	
		DstAilObjectId - Possible values returned																
M_BACKGROUND_IMAGE			Specifies to set the background image of a 3D display, or to copy the background image of a 3D display to an image buffer.															
	3D display identifier from which to copy		Specifies the identifier of the 3D display whose background to copy. The 3D display must have been previously allocated using M3ddispAlloc(). An error is generated if the 3D display does not have a background image.	c		e	f	g	a		i		j	k	l	m	n	b
		Image buffer identifier	Specifies the identifier of the 8-bit unsigned image buffer in which to copy the background image. The image buffer must have the same width, height, and number of bands as the background image of the 3D display. You can determine the size and number of bands of the current background image buffer using M3ddispInquire() with M_BACKGROUND_IMAGE_SIZE_....	c		e	f	g	a		i		j	k	l	m	n	b

Image buffer identifier from which to copy	Specifies the identifier of the 1- or 3-band, 8-bit unsigned image buffer to copy to the background of the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier	Specifies the identifier of the 3D display to which the image buffer is copied. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER	Specifies to copy the resulting image of the current render to the destination buffer or container. The render is guaranteed to be up to date when M_UPDATE is set to M_ENABLE. Otherwise, the image that is currently on the screen will be copied.	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier from which to copy	Specifies the identifier of the 3D display whose current render to copy. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
Container identifier	Specifies the identifier of the Aurora Imaging Library container in which to copy. The result of this will be a container with one component (M_COMPONENT_INTENSITY with the appropriate size, band, and data type). If a component like this already exists in the container, it will be reused; otherwise, the container will be emptied and a new component will be allocated.	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier	Specifies the identifier of the Aurora Imaging Library image buffer in which to copy. It is recommended to copy to an 8-bit, 3-band, unsigned image buffer. If the destination is smaller in size than the source, the image is cropped to the size of the destination buffer. If the destination is larger in size than the source, exceeding areas of the buffer are unaffected.	c		e	f	g	a		i		j	k	l	m	n	b

M_ROTATION_AXIS	Specifies to set the auto-rotation axis of a 3D display, or to copy the auto-rotation axis of a 3D display to a 3D line geometry object. This is the axis that the view of the 3D display rotates around when M_AUTO_ROTATE is set to M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier from which to copy	Specifies the identifier of the 3D display whose auto-rotation axis to copy. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object identifier	Specifies the identifier of the 3D geometry object to define. The 3D geometry object is defined as an infinite line.	c		e	f	g	a		i		j	k	l	m	n	b
3D line geometry object identifier	Specifies the identifier of the 3D line geometry object used to set the 3D display's view. The 3D line geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and defined as a line.	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier	Specifies the identifier of the 3D display whose view to set. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATION_AXIS_CENTER	Specifies to set the center position of the auto-rotation axis of a 3D display, or to copy this position from a 3D display to a 3D point geometry object. This is the point that the view of the 3D display rotates around when M_AUTO_ROTATE is set to M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_INTEREST_POINT	Specifies to set the 3D display's rotation axis center point to its interest point.	c		e	f	g	a		i		j	k	l	m	n	b

3D display identifier	Specifies the identifier of the 3D display whose view to set. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier from which to copy	Specifies the identifier of the 3D display whose auto-rotation center point to copy. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
3D geometry object identifier	Specifies the identifier of the 3D geometry object in which to copy the position. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY The 3D geometry object is defined as a 3D point.	c		e	f	g	a		i		j	k	l	m	n	b
3D point geometry object identifier	Specifies the identifier of the 3D point geometry object used to set the 3D display's auto-rotation center point. The 3D point geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and defined as a point.	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier	Specifies the identifier of the 3D display whose view to set. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_MATRIX +	Specifies to copy the view's transformation matrix. If copied before the view of the 3D display is moved, you can use this matrix with **M3ddispCopy**() or M3ddispSetView() to restore the current position and orientation of the view. The transformation matrix of a view is relative to a view with the viewpoint at (0,0,1), the interest point at (0,0,0), and an up vector of (0,1,0). This can also be thought of as a direct top-down view, with the up vector aligned with the Y-axis.	c		e	f	g	a		i		j	k	l	m	n	b

M_IDENTITY_MATRIX	Specifies to set the 3D display's view to the identity matrix. This can be used to reset the view's orientation.	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier	Specifies the identifier of the 3D display whose view to set. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier from which to copy	Specifies the identifier of a 3D display from which to copy the view's orientation. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
Transformation matrix object identifier	Specifies the identifier of the transformation matrix object in which to copy the view's orientation. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b
Transformation matrix object identifier	Specifies the identifier of the transformation matrix object used to set the orientation of the 3D display's view. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b
3D display identifier	Specifies the identifier of the 3D display of which to set the view's orientation. The 3D display must have been previously allocated using M3ddispAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_VIEW_MATRIX.

You can add the following value to the above-mentioned value to specify to use the current viewpoint as the starting point for the transformation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Specifies the starting position																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_COMPOSE_WITH_CURRENT	Specifies to use the current viewpoint as the starting point for the transformation. This setting can only be used when copying a transformation matrix object to a 3D display.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispCopyResult

Synopsis

Copy a group of results from a picking area result buffer to an image buffer.

Syntax

```
void M3ddispCopyResult(
    AIL_ID Result3ddispId, //in
    AIL_INT64 SrcIndex,    //in
    AIL_ID DstImageBufId,  //out
    AIL_INT64 DstIndex,    //in
    AIL_INT64 CopyType,    //in
    AIL_INT64 ControlFlag  //in
)
```

Description

This function copies results from a 3D display picking area result buffer into an image buffer. You can use the resulting image buffer as a mask image.

Parameters

Result3ddispId

Specifies the identifier of the 3D display picking area result buffer from which to copy. The result buffer must have been allocated using M3ddispAllocResult() with M_PICKING_AREA_RESULT, and must contain the results of a call to M3ddispPickRect().

SrcIndex

Specifies the index of the picking area result to copy. The specified index must correspond to a point cloud graphic.

> Note that point cloud graphics generated from a depth map image buffer are not supported.

DstImageBufId

Specifies the identifier of the destination image buffer in which to copy.

The image buffer must be 1-band and must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

The image buffer must have the same dimensions as the range component of the container from which the point cloud graphic was generated. You can inquire the component's size using MbufInquireContainer() with M_COMPONENT_RANGE, M_SIZE_X, and M_SIZE_Y.

DstIndex

Reserved for future expansion and must be set to **M_DEFAULT**.

CopyType

Specifies the type of copy operation to perform.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Specifies the type of copy operation																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_MASK_IMAGE	Specifies to create a mask image whose pixels are non-zero for picked points within the picking region, and zero otherwise. Qualifying points are those of the point cloud graphic that were within the picking region when the graphic was picked. This operation maps the indices of the point cloud graphic's sub-elements (M3ddispGetResult() with M_GRAPHIC_SUB_INDICES) to corresponding pixel locations in the destination image buffer, which are assigned the maximum buffer value. All other pixels in the buffer are set to 0.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispFree

Synopsis

Free a 3D display, picking context, or picking result buffer.

Syntax

```
void M3ddispFree(
    AIL_ID Disp3dId //in
)
```

Description

This function deallocates a 3D display, picking context, or picking result buffer previously allocated with M3ddispAlloc() or M3ddispAllocResult().

If the 3D display was shown in a window using M3ddispSelect(), **M3ddispFree**() also closes the associated window; if the 3D display was shown in a window using M3ddispSelectWindow(), the associated window is left open but it is left blank.

All 3D displays, picking contexts, and picking result buffers allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

> An error will be generated if you attempt to free the internal 3D graphics list associated with a 3D display.

Parameter

Disp3dId

Specifies the identifier of a 3D display, picking context, or picking result buffer previously allocated with M3ddispAlloc() or M3ddispAllocResult().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispGetHookInfo

Synopsis

Get information about a 3D display hook event.

Syntax

```
AIL_INT M3ddispGetHookInfo(
    AIL_ID EventId,     //in
    AIL_INT64 InfoType, //in
    void *UserVarPtr    //out
)
```

Description

This function allows you to get information about the event that caused the hook-handler function to be called. **M3ddispGetHookInfo**() should only be called within the scope of a 3D display hook-handler function (see M3ddispHookFunction()).

Parameters

EventId

Specifies the 3D display event identifier received by the hook-handler function (see M3ddispHookFunction()).

InfoType

Specifies the type of information about the event to return.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_ID
- AIL_INT

Specifies the address in which to write the requested information.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InfoType** parameter and possible values returned to the **UserVarPtr** parameter.

- For retrieving information about a general 3D display hook event
- For retrieving information regarding mouse events
- For retrieving information regarding keyboard events
- For retrieving information regarding either a mouse or keyboard event
- For retrieving information regarding frametime events

Regardless of the 3D display event that caused the hook-handler function to be called, the InfoType parameter can be set to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information about a general 3D display hook event																	
InfoType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_DISPLAY		Retrieves the Aurora Imaging Library identifier of the 3D display that generated the event.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
	3D display identifier	Specifies the Aurora Imaging Library identifier of the 3D display that generated the event.	c		e	f	g	a		i		j	k	l	m	n	b

If the hook-handler function was called due to an M_MOUSE_... event type, the InfoType parameter can be set to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information regarding mouse events																	
InfoType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_MOUSE_POSITION_X		Retrieves the X-position of the cursor, in display coordinates.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the X-position of the cursor, in display coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_POSITION_Y		Retrieves the Y-position of the cursor, in display coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the Y-position of the cursor, in display coordinates.	c		e	f	g	a		i		j	k	l	m	n	b

M_MOUSE_WHEEL_VALUE	Retrieves the value for the mouse wheel's rotation. Note, this value can only be retrieved if the hook-handler function was called due to the M_MOUSE_WHEEL event.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the value for the mouse wheel's rotation. A positive value indicates that the wheel was rotated forward, away from the user; a negative value indicates that the wheel was rotate backward, toward the user.	c		e	f	g	a		i		j	k	l	m	n	b

If the hook-handler function was called due to an M_KEY_... event type, the InfoType parameter can be set to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information regarding keyboard events																	
InfoType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																	
M_AIL_KEY_VALUE	Retrieves the Aurora Imaging Library operating system-independent key code for the keyboard key that triggered the event. Note, this value can only be retrieved if the hook-handler function was called due to the M_KEY_DOWN, M_KEY_UP, or M_KEY_CHAR event.																
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ADD	Specifies the Add key (+).	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_DOWN	Specifies the Down Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_LEFT	Specifies the Left Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_RIGHT	Specifies the Right Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_UP	Specifies the Up Arrow key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_BACK	Specifies the Back key.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR	Specifies the Clear key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL	Specifies the Period key located on your number pad.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE	Specifies the Delete key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE	Specifies the Divide key (/).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END	Specifies the End key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC	Specifies the Esc key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE	Specifies the Execute key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n*	Specifies the *n* key, where *n* is a value from 1 to 24.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME	Specifies the Home key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT	Specifies the Insert key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY	Specifies the Multiply key (*).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n*	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK	Specifies the Num Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n*	Specifies the *n* key, where *n* is a value from 0 to 9.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN	Specifies the Pagedown key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP	Specifies the Pageup key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE	Specifies the Pause key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN	Specifies the Return key (Enter).	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK	Specifies the Scroll Lock key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR	Specifies the Separator key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE	Specifies the Space key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT	Specifies the Subtract key (-).	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB	Specifies the Tab key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx*	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_VALUE		Retrieves the operating system's key code for the keyboard key that triggered the event.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the key code.	c		e	f	g	a		i		j	k	l	m	n	b

If the hook-handler function was called due to either an M_KEY_... or M_MOUSE_... event type, the InfoType parameter can be set to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information regarding either a mouse or keyboard event																	
InfoType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_COMBINATION_KEYS		Retrieves which combination keys or mouse buttons triggered the event. Note that a combination of the values below can be returned. Bitwise operators must be used to establish the specific keys or buttons pressed.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ALT	Specifies the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CTRL	Specifies the Ctrl key.	c		e	f	g	a		i		j	k	l	m	n	b

M_KEY_SHIFT	Specifies the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_WIN	Specifies the Windows key.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_LEFT_BUTTON	Specifies the left mouse button.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_MIDDLE_BUTTON	Specifies the middle mouse button.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_RIGHT_BUTTON	Specifies the right mouse button.	c		e	f	g	a		i		j	k	l	m	n	b

If the hook-handler function was called due to an M_FRAMETIME_... event type, the InfoType parameter can be set to one of the values below.

> Note, the frametime is the time it took to render a frame.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information regarding frametime events																	
InfoType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_FRAMETIME_AVERAGE	Retrieves the average frametime in seconds, over the number of frames specified by M_FRAMETIME_AVERAGE_SMOOTHING.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the frametime.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_AVERAGE_SMOOTHING	Retrieves the number of frames used when calculating M_FRAMETIME_AVERAGE.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the number of frames.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_LAST	Retrieves the frametime of the last frame in seconds.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies the frametime.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is **M_NULL** if successful. If the operation fails, a non-null (!**M_NULL**) value is returned.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispGetResult

Synopsis

Get the specified type of result from a picking result buffer.

Syntax

```
AIL_DOUBLE M3ddispGetResult(
    AIL_ID Result3ddispId, //in
    AIL_INT64 ResultIndex, //in
    AIL_INT64 ResultType,  //in
    void *ResultArrayPtr   //out
)
```

Description

This function retrieves results of the specified type from a picking result buffer. Results are available after calling M3ddispPick() or M3ddispPickRect().

Parameters

Result3ddispId

Specifies the identifier of the picking result buffer, previously allocated using M3ddispAllocResult() with M_PICKING_RESULT or M_PICKING_AREA_RESULT.

ResultIndex

Specifies the index of the result to get. Set this parameter to one of the values below:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying which result to retrieve																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_PICKING_INDEX(AIL_INT PickingIndex)	Specifies to retrieve results for a detected 3D graphic at the specified index. For an M_PICKING_RESULT buffer, the sole result (if detected) is stored at index 0.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PickingIndex This parameter must be set to the following:															

			Value >= 0	Specifies the index of the pick result.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to retrieve a general result from a picking result buffer.				c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result to get.

See the Parameter associations section for possible values that can be specified.

ResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- AIL_DOUBLE
- AIL_FLOAT
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the results. Since **M3ddispGetResult**() also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ResultType** parameter and possible values returned to the **ResultArrayPtr** parameter.

- For retrieving general results from a picking result buffer
- For retrieving results of a particular graphic that was picked

To retrieve general results from a picking result buffer, set the ResultIndex parameter to M_GENERAL. In this case, you can set the ResultType parameter to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving general results from a picking result buffer																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultArrayPtr - Possible values returned																

M_NUMBER +	Retrieves the number of 3D graphics in the picking result buffer. Note that an M_PICKING_RESULT buffer will store either 0 or 1 detected 3D graphics. This result is always available.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET_POSITION_DISPLAY_X +	Retrieves the X-pixel coordinate used for the pick operation. This result is always available after an M3ddispPick() operation. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_TARGET_POSITION_DISPLAY_Y +	Retrieves the Y-pixel coordinate used for the pick operation. This result is always available after an M3ddispPick() operation. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve results of a picked graphic, set the ResultIndex parameter to M_PICKING_INDEX(). The ResultIndex parameter must be set to M_PICKING_INDEX(0) when retrieving results from an M_PICKING_RESULT buffer. Set the ResultType parameter to one of the following values:

> Note, the following results are only available if the pick operation detected a 3D graphic.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving results of a particular graphic that was picked																
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																

M_CONTAINER_ID +	Retrieves the Aurora Imaging Library identifier of the container or depth map image buffer linked to the detected 3D graphic. This result is only available if the pick operation detected a point cloud graphic, and that point cloud graphic was not created with M_NO_LINK.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
Container identifier	Specifies the Aurora Imaging Library identifier of the container linked to the 3D graphic.	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier	Specifies the Aurora Imaging Library identifier of the depth map image buffer linked to the 3D graphic.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISTANCE +	Retrieves the distance from the picked position to the 3D display's viewpoint. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_FACE +	Retrieves the index of the picked face. This result is only available if the pick operation detected a box or a meshed point cloud graphic. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_ID +	Retrieves the Aurora Imaging Library identifier of the 3D graphics list that contains the detected 3D graphic.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b

3D graphics list identifier	Specifies the Aurora Imaging Library identifier of the 3D graphics list containing the 3D graphic that was detected by the pick operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SUB_INDEX +	Retrieves the index of the picked sub-element (for example, the index of the picked dot in the detected dots graphic). This result is only available if the pick operation detected a dots or lines graphic. Note that lines graphics must not have their M_APPEARANCE set to M_POINTS. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SUB_INDEX_NUMBER +	Retrieves the number of graphic sub-elements were picked for the given graphic. This result is only available if the pick operation detected a dots, lines, or point cloud graphic. This result is only available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the number of sub-elements that were picked.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_SUB_INDICES +	Retrieves an array of the indices of the specified graphic's sub-elements within the picking area. For example, for a point cloud graphic, its sub-elements are those points that were within the picking region during the call to M3ddispPickRect(). This result is only available if the pick operation detected a dots, lines, or point cloud graphic. This result is only available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b

	ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the indices of the sub-elements within the area.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE +		Retrieves the type of the detected 3D graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_ARC	Specifies an arc graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_ARC_FILL	Specifies a filled arc graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_AXIS	Specifies an axis graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_BOX	Specifies a box graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_CYLINDER	Specifies a cylinder graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_DOTS	Specifies dots graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_GRID	Specifies a grid graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINE	Specifies a line graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINES	Specifies a lines graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_PLANE	Specifies a plane graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POINT_CLOUD	Specifies a point cloud graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POLYGON	Specifies a polygon graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_RECT	Specifies a rectangle graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_SPHERE	Specifies a sphere graphic.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_TEXT	Specifies text graphic.	c		e	f	g	a		i		j	k	l	m	n	b

M_LABEL +	Retrieves the label of the detected 3D graphic.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_PICKED_POINT_PIXEL_X +	Retrieves the X-pixel coordinate of the picked point in the range component or depth map. This result is only available if the pick operation detected a point cloud graphic. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_PICKED_POINT_PIXEL_Y +	Retrieves the Y-pixel coordinate of the picked point in the range component or depth map. This result is only available if the pick operation detected a point cloud graphic. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

M_PICKED_POSITION_3D_X +	Retrieves the X-position where the ray intersected with the 3D graphic in world coordinates. For a point cloud or dots graphic, this is the X-position of the point in the 3D graphic. Note that this result can differ (very slightly) from the value stored in the range component of a point cloud or dots graphic because, a point might not line up with a specific pixel on the display. For a point cloud graphic, you can use MbufGet() to copy the values in the range component into an array, and then retrieve the stored X-position of the picked point from this array, using M_PICKED_POINT_PIXEL_X and M_PICKED_POINT_PIXEL_Y and 0 as the indices. For a dots graphic, you can use M_GRAPHIC_SUB_INDEX to retrieve the X-position of the picked point from the array that you used to create the dots graphics. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_PICKED_POSITION_3D_Y +	Retrieves the Y-position where the ray intersected with the 3D graphic in world coordinates. For a point cloud or dots graphic, this is the Y-position of the point in the 3D graphic. Note that this result can differ (very slightly) from the value stored in the range component of a point cloud or dots graphic because, a point might not line up with a specific pixel on the display. For a point cloud graphic, you can use MbufGet() to copy the values in the range component into an array, and then retrieve the stored Y-position of the picked point from this array, using M_PICKED_POINT_PIXEL_X and M_PICKED_POINT_PIXEL_Y and 0 as the indices. For a dots graphic, you can use M_GRAPHIC_SUB_INDEX to retrieve the index of the picked point from the array that you used to create the dots graphics. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_PICKED_POSITION_3D_Z +	Retrieves the Z-position where the ray intersected with the 3D graphic in world coordinates. For a point cloud or dots graphic, this is the Z-position of the point in the 3D graphic. Note that this result can differ (very slightly) from the value stored in the range component of a point cloud or dots graphic because, a point might not line up with a specific pixel on the display. For a point cloud graphic, you can use MbufGet() to copy the values in the range component into an array, and then retrieve the stored Z-position of the picked point from this array, using M_PICKED_POINT_PIXEL_X and M_PICKED_POINT_PIXEL_Y and 0 as the indices. For a dots graphic, you can use M_GRAPHIC_SUB_INDEX to retrieve the index of the picked point from the array that you used to create the dots graphics. This result is not available for an M_PICKING_AREA_RESULT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_PICKED_POSITION_3D_X; M_PICKED_POSITION_3D_Y; M_PICKED_POSITION_3D_Z.

You can add the following value to the above-mentioned values to get the position relative to the 3D graphic's root node.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a position relative to the 3D graphics list's root node																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_RELATIVE_TO_ROOT	Specifies to get the picked position relative to the 3D graphics list's root node.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For retrieving general results from a picking result buffer, For retrieving results of a particular graphic that was picked.

You can add the following value to the above-mentioned values to determine whether a result is available.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining whether results are available

ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_AVAILABLE	Retrieves whether the requested result type is available for retrieval.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the requested result type is not available.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the requested result type is available.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For retrieving general results from a picking result buffer, For retrieving results of a particular graphic that was picked.

You can add one of the following values to the above-mentioned values to cast the requested results to the required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested results to an *AIL_DOUBLE*.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT	Casts the requested results to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested results to an *AIL_ID*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested results to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

M_TYPE_AIL_INT32	Casts the requested results to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested results to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_DOUBLE*. If the requested information does not fit into an *AIL_DOUBLE*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispGetView

Synopsis

Get information about the 3D display's view.

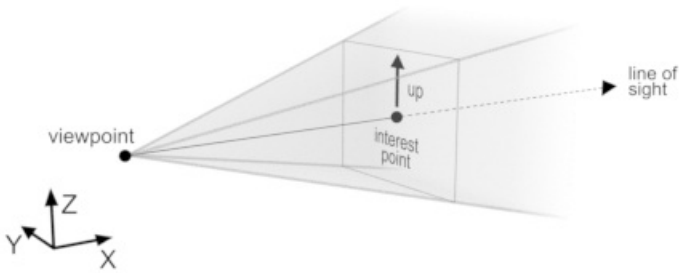
Syntax

```
void M3ddispGetView(
    AIL_ID Disp3dId,       //in
    AIL_INT64 Mode,        //in
    AIL_DOUBLE *Param1Ptr, //out
    AIL_DOUBLE *Param2Ptr, //out
    AIL_DOUBLE *Param3Ptr, //out
    AIL_INT64 ControlFlag  //in
)
```

Description

This function gets information about the current view of the specified 3D display.

All settings of the view of a 3D display are based on the viewpoint, interest point, and/or up vector, as shown in the image below:



You can get the position of the viewpoint using M_VIEWPOINT, the position of the interest point using M_INTEREST_POINT and the direction of the up-vector using M_UP_VECTOR. The results of all other modes are derived from these modes. For example, the M_ROLL setting of the view is derived from the M_UP_VECTOR setting.

All results are returned with respect to the working coordinate system of the 3D display.

Parameters

Disp3dId

Specifies the identifier of the 3D display.

Mode

Specifies which aspect of the view to inquire.

See the Parameter associations section for possible values that can be specified.

Param1Ptr

Specifies the address of the variable in which to write either the X-coordinate, distance, azimuth, elevation, or roll.

See the Parameter associations section for possible values that can be returned.

Param2Ptr

Specifies the address of the variable in which to write either the Y-coordinate or elevation.

Set this parameter to **M_NULL** if not used.

See the Parameter associations section for possible values that can be returned.

Param3Ptr

Specifies the address of the variable in which to write either the Z-coordinate or roll.

Set this parameter to **M_NULL** if not used.

See the Parameter associations section for possible values that can be returned.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

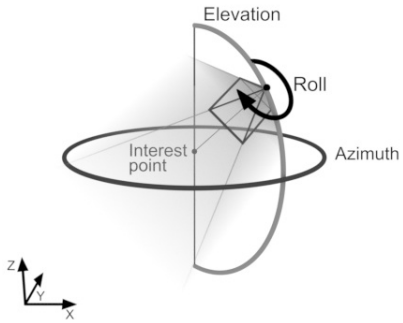
Parameter associations

The table below lists possible values for the **Mode** parameter and possible values returned to the **Param1Ptr**, **Param2Ptr**, and **Param3Ptr** parameters.

- For determining the position and orientation of the viewpoint.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining the position and orientation of the viewpoint.																	
Mode		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	Param1Ptr - Possible values returned																
	Param2Ptr - Possible values returned																
	Param3Ptr - Possible values returned																

M_AZIM_ELEV_ROLL		Gets the azimuth and elevation of the viewpoint relative to the interest point, and the roll of the view. These are the viewpoint position and up vector, described in a spherical coordinate system (additionally described by M_DISTANCE). Azimuth is the angle of the viewpoint relative to the interest point on the XY-plane. An azimuth of 0 corresponds to a view where the viewpoint and interest point are parallel to the X-axis on the XY-plane, with the viewpoint on the positive side. Increasing values for the azimuth correspond to rotating the view counter-clockwise around the interest point. Elevation is the angle of the viewpoint relative to the interest point on the XZ-plane. An elevation of 0 corresponds to a view where the viewpoint and interest point are parallel to the X-axis on the XZ-plane, with the viewpoint on the positive side. Increasing values for the elevation correspond to rotating the view counter-clockwise around the interest point in the XZ-plane. Roll is the angle that describes how much the view is rotated to the right. A roll of 0 corresponds to a view for which up is positive on the Z-axis. 														
	Param1Ptr	Returns the azimuth of the view, specified in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2Ptr	Returns the elevation of the view, specified in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	Param3Ptr	Returns the roll of the view, specified in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_AZIMUTH	Gets the azimuth of the viewpoint relative to the interest point. Azimuth is the angle of the viewpoint relative to the interest point on the XY-plane. An azimuth of 0 corresponds to a view where the viewpoint and interest point are parallel to the X-axis on the XY-plane, with the viewpoint on the positive side. Increasing values for the azimuth correspond to rotating the view counter-clockwise around the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the azimuth of the viewpoint relative to the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISTANCE	Gets the distance of the viewpoint from the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the distance of the viewpoint from the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELEVATION	Gets the elevation of the viewpoint relative to the interest point. Elevation is the angle of the viewpoint relative to the interest point on the XZ-plane. An elevation of 0 corresponds to a view where the viewpoint and interest point are parallel to the X-axis on the XZ-plane, with the viewpoint on the positive side. Increasing values for the elevation correspond to rotating the view counter-clockwise around the interest point in the XZ-plane.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the elevation of the viewpoint relative to the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTEREST_POINT	Gets the position of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the X-coordinate of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param2Ptr	Returns the Y-coordinate of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param3Ptr	Returns the Z-coordinate of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b

M_ROLL	Gets the roll of the view. This is an angle which defines what direction the top of the view will be pointing, with 0 degrees specifying that the view will be perfectly level. As this value increases, the view rolls to the right.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the roll of the view, specified in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_UP_VECTOR	Gets the vector which defines what direction the top of the view is pointing. Note that this setting will always return a unit vector that is orthogonal to the current orientation of the view. The value returned by this function will therefore typically be different from the up vector that you might have previously specified using M3ddispSetView() with M_UP_VECTOR.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the X-component of the up vector.	c		e	f	g	a		i		j	k	l	m	n	b
Param2Ptr	Returns the Y-component of the up vector.	c		e	f	g	a		i		j	k	l	m	n	b
Param3Ptr	Returns the Z-component of the up vector.	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_ORIENTATION	Gets the unit vector that describes the direction from the viewpoint to the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the X-component of the view orientation.	c		e	f	g	a		i		j	k	l	m	n	b
Param2Ptr	Returns the Y-component of the view orientation.	c		e	f	g	a		i		j	k	l	m	n	b
Param3Ptr	Returns the Z-component of the view orientation.	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEWPOINT	Gets the position of the viewpoint.	c		e	f	g	a		i		j	k	l	m	n	b
Param1Ptr	Returns the X-coordinate of the position of the viewpoint.	c		e	f	g	a		i		j	k	l	m	n	b
Param2Ptr	Returns the Y-coordinate of the position of the viewpoint.	c		e	f	g	a		i		j	k	l	m	n	b

	Param3Ptr	Returns the Z-coordinate of the position of the viewpoint.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispHookFunction

Synopsis

Hook a function to a 3D display event.

Syntax

```
void M3ddispHookFunction(
    AIL_ID Disp3dId,                            //out
    AIL_INT64 HookType,                         //in
    AIL_3DDISP_HOOK_FUNCTION_PTR HookHandlerPtr, //in
    void *UserDataPtr                           //in-out
)
```

Description

This function allows you to attach or detach a user-defined function to a specified 3D display event. Once a hook-handler function is defined and hooked to an event, it is automatically called when the event occurs.

Note that if a mouse or keyboard event is also associated with user interaction, the hook-handler function is called before processing the user interaction. You can use M3ddispSetHookInfo() with M_CONSUMED set to M_TRUE to prevent the user interaction from being processed entirely.

You can hook more than one function to an event by making separate calls to **M3ddispHookFunction**() for each function that you want to hook. Aurora Imaging Library automatically chains and keeps an internal list of all these hooked functions. When a function is hooked, this new function is added to the end of the list. When the event happens, all user-defined functions in the list will be executed in the same order that they were hooked to the event. You can also remove any function from the list; in this case, Aurora Imaging Library preserves the order of the remaining functions in the list.

Hook functions are called from the 3D display's internal thread; as such, any expensive or blocking task can freeze the 3D display. All 3D display and 3D graphics functions are available from within the callback and will not race with anything for this 3D display; however, controlling or inquiring another 3D display might be a blocking call in some cases (in particular, all calls to M3ddispInquire() and M3ddispGetView() are blocking calls). You should not use hook functions with two 3D displays that might try to inquire each other at the same time. For example, 3D display A should not be trying to get the M_VIEWPOINT of 3D display B while 3D display B is trying to copy the M_VIEW_MATRIX of 3D display A (this would deadlock the two displays).

Parameters

Disp3dId

Specifies the identifier of the target 3D display.

HookType

Specifies the 3D display event type. This parameter can be set to the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 3D display event type																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_ENTER_SAFE_MODE +	Calls the hook-handler function when an unrecoverable error is encountered and the 3D display enters safe mode. When in safe mode, all calls to the 3D Display module will generate an error (except M3ddispGetHookInfo() and M3ddispFree()). When this event occurs, you should save your work and restart the application. Note that although all 3D displays will enter safe mode, only the 3D display that encountered the error will generate this event.	c		e	f	g	a		i		j	k	l	m	n	b
M_FOV_CHANGE +	Calls the hook-handler function each time the 3D display's field of view (FoV) changes. This includes setting the projection mode (M_PROJECTION_MODE), setting the FoV angle in M_PERSPECTIVE or M_PARALLEL, setting the parallel scale in M_PARALLEL_PRECISE, changing the distance in M_PARALLEL, or changing the window size. Note that the hook function will typically be called right before the next render after the field of view changes. To guarantee that the 3D display is up to date, you can force a render, using M3ddispControl() with M_UPDATE set to M_NOW.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_HIGH +	Calls the hook-handler function when the frametime goes above M_FRAMETIME_THRESHOLD_HIGH. Note that this event only occurs when the frametime first passes the threshold. This is reset whenever a new function is hooked to this event, any control related to frametime averaging is changed, or the frametime goes below M_FRAMETIME_THRESHOLD_LOW.	c		e	f	g	a		i		j	k	l	m	n	b

M_FRAMETIME_LOW +	Calls the hook-handler function when the frametime goes below M_FRAMETIME_THRESHOLD_LOW. Note that this event only occurs when the frametime first passes the threshold. This is reset whenever a new function is hooked to this event, any control related to frametime averaging is changed, or the frametime goes above M_FRAMETIME_THRESHOLD_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CHAR +	Calls the hook-handler function each time the user presses a character key, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_DOWN +	Calls the hook-handler function each time the user presses a keyboard key, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_UP +	Calls the hook-handler function each time the user releases a keyboard key, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_LEFT_BUTTON_DOWN +	Calls the hook-handler function each time the user clicks the left mouse button, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_LEFT_BUTTON_UP +	Calls the hook-handler function each time the user releases the left mouse button, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_LEFT_DOUBLE_CLICK +	Calls the hook-handler function each time the user double-clicks the left mouse button, while the mouse cursor is in the 3D display. Note that M_MOUSE_LEFT_DOUBLE_CLICK might not work with Aurora Imaging Web Library displays.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_MIDDLE_BUTTON_DOWN +	Calls the hook-handler function each time the user clicks the middle mouse button, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b

M_MOUSE_MIDDLE_BUTTON_UP +	Calls the hook-handler function each time the user releases the middle mouse button, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_MOVE +	Calls the hook-handler function each time the mouse cursor moves, while the moue cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_RIGHT_BUTTON_DOWN +	Calls the hook-handler function each time the user clicks the right mouse button, while the mouse cursor is in the display.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_RIGHT_BUTTON_UP +	Calls the hook-handler function each time the user releases the right mouse button, while the mouse cursor is in the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_WHEEL +	Calls the hook-handler function each time the user scrolls the mouse wheel (up or down), while the mouse cursor is in the display.	c		e	f	g	a		i		j	k	l	m	n	b

M_VIEW_CHANGE +	Calls the hook-handler function each time the view of the 3D display changes. This event occurs when the position or orientation of the view is modified, either using M3ddispSetView(), or through direct user interaction with the 3D display using the keyboard and mouse. If several changes are made to the view sequentially, there is no guarantee that this event will be generated for every intermediate change; if there is a single callback, it will be associated with the final view change. This event is not generated for a field of view change (such as resizing the window). See M_FOV_CHANGE instead. M_VIEW_CHANGE is not recursive; calling M3ddispSetView() for this 3D display from within the callback will not generate more events. This event can be triggered even if no actual modification occurred. For example, if you attempt to set the view to the same position and orientation as the current view (using M3ddispCopy() with M_VIEW_MATRIX to store the current transformation matrix of the view, and then setting the view to match that transformation matrix, using M3ddispCopy() or M3ddispSetView() with M_VIEW_MATRIX), there is a chance the hook function will be called again, even though the view did not visibly change. Note that when calling M3ddispSetView(), it is not guaranteed that the view change event will have been generated yet when the function returns. If you need to guarantee that the view change event has been generated, use M3ddispControl() with M_SYNCHRONIZE after the call to M3ddispSetView().	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_RESIZED +	Calls the hook-handler function each time the window is resized.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For specifying the 3D display event type.

You can add the following value to the above-mentioned values to set whether to detach the hook-handler

function.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For detaching the hook-handler function																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_UNHOOK	Unhooks the specified function if hooked to the specified event. When you use M_UNHOOK, you must provide the same values for all the parameters as when you originally hooked the function.	c		e	f	g	a		i		j	k	l	m	n	b

HookHandlerPtr

Specifies the address of the function that should be called when the specified event occurs. The hook-handler function, pointed to by HookHandlerPtr, must be declared as follows:

```
AIL_INT MFTYPE HookHandler(
    AIL_INT HookType,
    AIL_ID EventId,
    void *UserDataPtr
)
```

Parameters

HookType

Type of event hooked.

EventId

Event identifier to pass to M3ddispGetHookInfo() when inquiring about the hooked event.

UserDataPtr

Data pointer that was passed to **M3ddispHookFunction**().

Upon successful completion, the hook-handler function should return **M_NULL**. Note, MFTYPE and AIL_3DDISP_HOOK_FUNCTION_PTR are reserved Aurora Imaging Library predefined types for functions and data pointers.

UserDataPtr

Specifies the address of the user data that you want to make available to the hook-handler function. This address is passed to the hook-handler function, through its UserDataPtr parameter, when the specified event occurs. Set this parameter to **M_DEFAULT** if not used.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispInquire

Synopsis

Inquire an Aurora Imaging Library 3D display or picking context setting.

Syntax

```
AIL_INT64 M3ddispInquire(
    AIL_ID Disp3dId,       //in
    AIL_INT64 InquireType, //in
    void *UserVarPtr       //out
)
```

Description

This function allows you to inquire the specified Aurora Imaging Library 3D display or picking context setting.

Parameters

Disp3dId

Specifies the identifier of the target 3D display or picking context, previously allocated using M3ddispAlloc().

InquireType

Specifies the type of 3D display or picking context setting to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]
- array of type AIL_TEXT_CHAR [optionally, in C++: a reference to an AIL_STRING]
- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT64

Specifies the address in which to write the requested information.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For inquiring keyboard and mouse controls
- For 3D displays
- For picking contexts

You can use the following InquireType to inquire about the keyboard and mouse bindings used to control of the 3D display.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring keyboard and mouse controls																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_ACTION_KEY_AUTO_ROTATE +	Inquires the keyboard keys that you can press to start and stop auto-rotation around the interest point. This action is equivalent to enabling or disabling auto-rotation using M_AUTO_ROTATE. SET															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_R +	Specifies the R key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORBIT_DOWN +		Inquires the keyboard keys that you can press to orbit the viewpoint down around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL and a negative value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORBIT_LEFT +		Inquires the keyboard keys that you can press to orbit the viewpoint left around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL and a negative value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ORBIT_RIGHT +		Inquires the keyboard keys that you can press to orbit the viewpoint right around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL and a positive value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORBIT_UP +		Inquires the keyboard keys that you can press to orbit the viewpoint up around the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL and a positive value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_BOTTOM_TILTED +		Inquires the keyboard keys that you can press to set the orientation of the viewpoint so that the view faces the bottom-front of the interest point. This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_BOTTOM_TILTED. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_1 +	Specifies the 1 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_BOTTOM_VIEW +		Inquires the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly below the interest point (parallel to the Z-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_BOTTOM_VIEW. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_2 +	Specifies the 2 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_FRONT_VIEW +	Inquires the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly in front of the interest point (parallel to the Y-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_FRONT_VIEW. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_5 +	Specifies the 5 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_LEFT_VIEW +		Inquires the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly to the left of the interest point (parallel to the X-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_LEFT_VIEW. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_4 +	Specifies the 4 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_REAR_VIEW +		Inquires the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly behind the interest point (parallel to the Y-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_REAR_VIEW. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_3 +	Specifies the 3 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_RIGHT_VIEW +		Inquires the keyboard keys that you can press to set the orientation of the viewpoint so that it is directly to the right of the interest point (parallel to the X-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_RIGHT_VIEW. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_6 +	Specifies the 6 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_TOP_TILTED +		Inquires the keyboard keys that you can press to set orientation of the viewpoint so that the view faces the top-front of the interest point. This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_TOP_TILTED. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_7 +	Specifies the 7 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ORIENTATION_TOP_VIEW +		Inquires the keyboard keys that you can press to set orientation of the viewpoint so that it is directly above the interest point (parallel to the Z-axis of the working coordinate system of the 3D display). This action is equivalent to using M3ddispSetView() with M_VIEW_ORIENTATION and M_TOP_VIEW. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_8 +	Specifies the 8 key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_RESET +		Inquires the keyboard keys that you can press to reset the view to the default position and orientation. This action is equivalent to using M3ddispSetView() with M_AUTO and M_TOP_TILTED. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ROLL_LEFT +		Inquires the keyboard keys that you can press to roll the view left. This action is equivalent to using M3ddispSetView() with M_ROLL + M_COMPOSE_WITH_CURRENT and a negative value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT + M_KEY_ALT	Specifies the Left Arrow key with the Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ROLL_RIGHT +		Inquires the keyboard keys that you can press to roll the view right. This action is equivalent to using M3ddispSetView() with M_ROLL + M_COMPOSE_WITH_CURRENT and a positive value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT + M_KEY_ALT	Specifies the Right Arrow key with the Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_BACKWARD +		Inquires the keyboard keys that you can press to move the view backward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN + M_KEY_ALT	Specifies the Down Arrow key with the Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_DOWN +		Inquires the keyboard keys that you can press to move the view downward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN + M_KEY_SHIFT	Specifies the Down Arrow key with the Shift key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_FORWARD +		Inquires the keyboard keys that you can press to move the view forward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_UP + M_KEY_ALT	Specifies the Up Arrow key with the Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_LEFT +		Inquires the keyboard keys that you can press to move the view left relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT + M_KEY_SHIFT	Specifies the Left Arrow key with the Shift key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_TRANSLATE_RIGHT +		Inquires the keyboard keys that you can press to move the view right relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT + M_KEY_SHIFT	Specifies the Right Arrow key with the Shift key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TRANSLATE_UP +		Inquires the keyboard keys that you can press to move the view upward relative to its current orientation. This action is equivalent to using M3ddispSetView() with M_TRANSLATE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP + M_KEY_SHIFT	Specifies the Up Arrow key with the Shift key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TURN_DOWN +		Inquires the keyboard keys that you can press to pan the view downward by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL + M_MOVE_INTEREST_POINT with a positive value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN + M_KEY_SHIFT + M_KEY_ALT	Specifies the Down Arrow key with the Shift key and Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_TURN_LEFT +		Inquires the keyboard keys that you can press to pan the view left by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL + M_MOVE_INTEREST_POINT with a positive value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT + M_KEY_SHIFT + M_KEY_ALT	Specifies the Left Arrow key with the Shift key and Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TURN_RIGHT +		Inquires the keyboard keys that you can press to pan the view right by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_HORIZONTAL + M_MOVE_INTEREST_POINT with a negative value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT + M_KEY_SHIFT + M_KEY_ALT	Specifies the Right Arrow key with the Shift key and Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_TURN_UP +		Inquires the keyboard keys that you can press to pan the view upward by moving the interest point. This action is equivalent to using M3ddispSetView() with M_ORBIT_VERTICAL + M_MOVE_INTEREST_POINT with a negative value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP + M_KEY_SHIFT + M_KEY_ALT	Specifies the Up Arrow key with the Shift key and Alt key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTION_KEY_ZOOM_IN +		Inquires the keyboard keys that you can press to move the viewpoint closer to the interest point. This action is equivalent to using M3ddispSetView() with M_DISTANCE and a value greater than 1.0. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_KEY_ZOOM_OUT +		Inquires the keyboard keys that you can press to move the viewpoint further from the interest point. This action is equivalent to using M3ddispSetView() with M_DISTANCE and a value between 0.0 and 1.0. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTION_MODIFIER_SPEED +		Inquires the keyboard key that you can use to modify the movement speed for all actions when using other keyboard commands. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ALT +	Specifies the Alt key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CTRL +	Specifies the CTRL key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SHIFT +	Specifies the Shift key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALTERNATE_SPEED_FACTOR +		Inquires by which factor the movement speed is multiplied when the key set using M_ACTION_MODIFIER_SPEED is held. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0.0	Specifies the factor by which to multiply the movement speed. The default value is 3.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_FUNCTION_KEY_CYCLE_APPEARANCES +		Inquires the keyboard keys that you can press to cycle through the given appearance modes for all point clouds currently being shown in the internal 3D graphics list of the 3D display. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CTRL + M_KEY_M +	Specifies the Ctrl key with the M key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_FUNCTION_KEY_POINTSIZE_DECREMENT +		Inquires the keyboard keys that you can press to decrement the point size for all point clouds currently being displayed. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_Z +	Specifies the Z key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_FUNCTION_KEY_POINTSIZE_INCREMENT +		Inquires the keyboard keys that you can press to increment the point size for all point clouds currently being displayed. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_X +	Specifies the X key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_FUNCTION_KEY_TOGGLE_LOD +		Inquires the keyboard keys that you can press to cycle through the given level of detail (LoD) modes for all point clouds currently being shown in the internal 3D graphics list of the 3D display. The LoD mode will cycle through the three modes (M_DISABLE, M_ENABLE_ON_ACTION, and M_ENABLE_ALWAYS). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ADD +	Specifies the Add key (+). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_DOWN +	Specifies the Down Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_LEFT +	Specifies the Left Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_RIGHT +	Specifies the Right Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_ARROW_UP +	Specifies the Up Arrow key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_BACK +	Specifies the Back key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_CLEAR +	Specifies the Clear key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DECIMAL +	Specifies the Period key located on your number pad. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DELETE +	Specifies the Delete key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_DIVIDE +	Specifies the Divide key (/). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_END +	Specifies the End key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_ESC +	Specifies the Esc key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_EXECUTE +	Specifies the Execute key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_F*n* +	Specifies the *n* key, where *n* is a value from 1 to 24. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_HOME +	Specifies the Home key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_INSERT +	Specifies the Insert key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_L +	Specifies the L key. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_MULTIPLY +	Specifies the Multiply key (*). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUM_LOCK +	Specifies the Num Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_NUMPAD*n* +	Specifies the *n* key, where *n* is a value from 0 to 9. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEDOWN +	Specifies the Pagedown key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAGEUP +	Specifies the Pageup key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_PAUSE +	Specifies the Pause key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_RETURN +	Specifies the Return key (Enter). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SCROLL_LOCK +	Specifies the Scroll Lock key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SEPARATOR +	Specifies the Separator key. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_KEY_SPACE +	Specifies the Space key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_SUBTRACT +	Specifies the Subtract key (-). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_TAB +	Specifies the Tab key. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_KEY_*xxx* +	Specifies the letter key *xxx*, where *xxx* can be a letter from A to Z. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NONE +	Specifies that there are no keyboard keys for this action setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEYBOARD_USE +		Inquires whether the user can interactively rotate, orbit, and scroll the view for the 3D display using the keyboard. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the keyboard to control the view is disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the keyboard to control the view is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_ROLL +		Inquires whether the user can interactively roll the view for the 3D display using the mouse. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to roll the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies that using the mouse to roll the view is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_ROTATION +	Inquires whether the user can interactively rotate the view for the 3D display using the mouse. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that using the mouse to rotate the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that using the mouse to rotate the view is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_TRANSLATION +	Inquires whether the user can interactively translate the view for the 3D display using the mouse. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that using the mouse to translate the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that using the mouse to translate the view is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_TRANSLATION_X +	Inquires whether the user can interactively translate the view for the 3D display in the X-direction using the mouse. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that using the mouse to translate the view in the X-direction is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to translate the view in the X-direction is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_TRANSLATION_Y +		Inquires whether the user can interactively translate the view for the 3D display in the Y-direction using the mouse. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to translate the view in the Y-direction is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to translate the view in the Y-direction is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_USE +		Inquires whether the user can interactively rotate, orbit, and scroll the view for the 3D display using the mouse. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to control the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to control the view is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOUSE_ZOOM +		Inquires whether the user can interactively zoom the view for the 3D display using the mouse. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that using the mouse to zoom the view is not enabled.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that using the mouse to zoom the view is enabled. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values of the UserVarPtr parameter listed in For inquiring keyboard and mouse controls. (exceptions might exist)

One of the following values might be combined with the above-mentioned values to specify an additional keyboard key.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying additional keyboard keys																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_KEY_ALT	Specifies the Alt key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_CTRL	Specifies the Ctrl key.	c		e	f	g	a		i		j	k	l	m	n	b
M_KEY_SHIFT	Specifies the Shift key.	c		e	f	g	a		i		j	k	l	m	n	b

The following InquireType and corresponding UserVarPtr parameter settings are available for 3D displays.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For 3D displays																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																

M_3D_GRAPHIC_LIST_ID +	Inquires the Aurora Imaging Library identifier of the internal 3D graphics list associated with this 3D display. Note that the 3D display is always associated with the same internal 3D graphics list, which is automatically allocated and freed at the same time as the 3D display. You should not attempt to free the internal graphics list. To display the contents of another 3D graphics list on a 3D display, you can copy the contents of the source 3D graphics list into the 3D display's internal 3D graphics list using M3dgraCopy(). You can also use M3ddispSelect() with M_ADD to add a 3D graphic to the internal graphics list or to add a graphics list to the 3D display context.													
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_ASSOCIATED_GRAPHIC_LIST_ID +	Inquires the 2D graphics list to associate with the 3D display. When a list is associated it must have no graphics set in world units and it cannot have any graphics with calibration set. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that no 2D graphics list is associated with the 3D display. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics list identifier	Specifies the identifier of the 2D graphics list to associate with the 3D display. The 2D graphics list must have been previously allocated using MgraAllocList().	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO_ROTATE +	Inquires whether to automatically rotate the view (both the viewpoint and interest point) around the auto-rotation axis. This gives the impression that the scene is rotating. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies to not auto-rotate the view. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to auto-rotate the view.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_COLOR +		Inquires the primary background color for the 3D display, used when M_BACKGROUND_MODE is set to M_SINGLE_COLOR or M_GRADIENT_VERTICAL. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_COLOR_GRADIENT +		Inquires the secondary background color for the 3D display, used when M_BACKGROUND_MODE specifies that the background is a gradient. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Byte-encoded RGB value	Specifies an encoded RGB value. To verify if the value is a byte-encoded RGB value, use the **M_IS_RGB888** macro. To retrieve the R, G, and B bands, use the **M_RGB888_R**, **M_RGB888_G**, and **M_RGB888_B** macros.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_IMAGE_SIZE_BAND +		Inquires the number of color bands of the background image buffer. This inquire type is only available if the 3D display has a background image buffer, set using M3ddispCopy() with M_BACKGROUND_IMAGE. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	1	Specifies the buffer has one band.	c		e	f	g	a		i		j	k	l	m	n	b

	3	Specifies the buffer has three bands.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_IMAGE_SIZE_X +		Inquires the width of the background image buffer. This inquire type is only available if the 3D display has a background image buffer, set using M3ddispCopy() with M_BACKGROUND_IMAGE. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the width of the buffer, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_IMAGE_SIZE_Y +		Inquires the height of the background image buffer. This inquire type is only available if the 3D display has a background image buffer, set using M3ddispCopy() with M_BACKGROUND_IMAGE. SET SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the height of the buffer, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_BACKGROUND_MODE +		Inquires whether the background of the 3D display is a solid color or a gradient. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_BACKGROUND_IMAGE	Specifies that the background is an image, set using M3ddispCopy() with M_BACKGROUND_IMAGE. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRADIENT_VERTICAL	Specifies that the background is a gradient. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SINGLE_COLOR	Specifies that the background is a solid color, specified by M_BACKGROUND_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_LUT_ID +	Inquires the identifier of the current LUT buffer associated with the 3D display's internal 3D graphics list. If the current LUT is predefined (default M_COLORMAP_TURBO), Aurora Imaging Library returns the type of predefined colormap to which the LUT is set (M_COLORMAP_...). If the current LUT is user-defined, the identifier of an internal copy of the user-defined LUT is returned. For a user-defined LUT that has an internal Aurora Imaging Library identifier, you must not free the associated buffer. You can, however, modify the buffer so that newly added point cloud graphics receive the modified LUT. Note that each point cloud graphic gets its own copy of the LUT associated with the 3D graphics list when it is created. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that the LUT buffer is set to M_NULL.	c		e	f	g	a		i		j	k	l	m	n	b
LUT buffer identifier	Specifies the Aurora Imaging Library identifier of the current LUT buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISPLAY_TYPE +	Inquires how the 3D display is presented. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_EXCLUSIVE	Specifies to present the 3D display full-screen, without a windowed border, in one of the Windows desktop screens. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WEB	Specifies to allocate a 3D display that is presented upon selection in one or more instances of an Aurora Imaging Web Library client application (for example, running in a compatible web browser). INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_WINDOWED	Specifies to allocate a 3D display that is presented upon selection in its own window on the Windows desktop screen(s). This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_FOV_HORIZONTAL_ANGLE +		Inquires the horizontal field of view of the 3D display. This is the angle which defines how much of the scene is shown within the width of the 3D display. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* < 180.0	Specifies the horizontal field of view, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_FOV_VERTICAL_ANGLE +		Inquires the vertical field of view of the 3D display. This is the angle which defines how much of the scene is shown within the height of the 3D display. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* < 180.0	Specifies the vertical field of view, in degrees. The default value is 30°.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_AVERAGE +		Inquires the average frametime in seconds, over the number of frames specified by M_FRAMETIME_AVERAGE_SMOOTHING.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the frametime in seconds.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_AVERAGE_SMOOTHING +		Inquires how many frames to use when calculating the average frametime returned using **M3ddispInquire**() with M_FRAMETIME_AVERAGE or M3ddispHookFunction() with M_FRAMETIME_HIGH or M_FRAMETIME_LOW. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0	Specifies the number of frames. The default value is 3.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_LAST +		Inquires the frametime of the last frame, in seconds; this is the time it took to render the frame. A low frametime indicates good GPU performance. A high frametime indicates poor GPU performance. If the frametime is too low, you either need to use a better GPU or reduce the complexity of your scene. For the 3D display to be usable interactively, you will typically want to target at least a frametime of 0.06 sec (equivalent to 15 fps). A frametime of 0.03 sec (equivalent to 30 fps) is preferable, and less than 0.016 sec (equivalent to 60 fps) is ideal. When grabbing continuously, if the frametime is higher than the time between grabs, some grabs will never be shown in the display.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the frametime, in seconds.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_THRESHOLD_HIGH +		Inquires the frametime above which to generate an M_FRAMETIME_HIGH event. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the frametime, in seconds. The default value is 0.066.	c		e	f	g	a		i		j	k	l	m	n	b
M_FRAMETIME_THRESHOLD_LOW +		Inquires the frametime below which to generate an M_FRAMETIME_LOW event. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the frametime, in seconds. The default value is 0.016.	c		e	f	g	a		i		j	k	l	m	n	b

M_GRAPHIC_LIST_OPACITY +	Inquires the opacity level of annotations generated from the 2D graphics list associated with the 3D display. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to use partial opacity. INFO	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* <= 100	Specifies the opacity (as a percentage), where 0 is completely transparent and 100 is completely opaque. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE_INPUT_MODE +	Inquires how the 3D display will handle user inputs. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_ASSOCIATED_GRAPHIC_LIST_2D	Specifies to use the mouse movements and keystrokes to draw graphics into the associated 2D graphics list (M_ASSOCIATED_GRAPHIC_LIST_ID).	c		e	f	g	a		i		j	k	l	m	n	b
M_NORMAL_3D	Specifies to use the mouse movements and keystrokes to manipulate the view of the 3D display. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOD_DEGRADATION_LEVEL +	Inquires the point/pixel density at which the display will choose a lower level of detail (LoD) for a given point cloud. A lower value will reduce the LoD more aggressively, improving performance at the cost of visible quality reduction. For example, a value below 1 will show lower LoDs even when the view is zoomed in, and values over 4 will rarely show lower LoDs. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/pixel density. The default value is 2.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOD_DEGRADE_MODE +	Inquires which level of detail (LoD) degradation mode to use. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to disable the LoD change. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Same as M_ENABLE_ON_ACTION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE_ALWAYS	Specifies to always show a degraded LoD.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE_ON_ACTION	Specifies to use one LoD lower when the user is interacting with the display. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LOD_TIME_TO_SHOW_DEGRADED +		Inquires how long to wait after an automatically degraded render, before rendering at the normally selected LoD. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies how long to wait, in seconds. The default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_PARALLEL_SCALE +		Inquires the parallel scale when using the M_PARALLEL_PRECISE projection mode. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the parallel scale. The default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROJECTION_MODE +		Inquires the mode of projection, which affects the appearance of 3D graphics shown in the 3D display. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_PARALLEL	Specifies to use parallel projection. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_PARALLEL_PRECISE	Specifies to use parallel projection, with a defined ratio between world units and pixel size. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_PERSPECTIVE	Specifies to use perspective projection. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATION_INDICATOR +		Inquires whether to show the rotation indicator in the center of the view. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to not show the rotation indicator.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to always show the rotation indicator.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE_ON_MOUSE_CLICK	Specifies to show the rotation indicator when the left mouse button is held down. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROTATION_SPEED +		Inquires how fast, and in what direction, the view rotates when M_AUTO_ROTATE is set to M_ENABLE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the auto-rotation speed, in degrees/sec. The default value is 20 degrees/sec.	c		e	f	g	a		i		j	k	l	m	n	b
M_SHOWN_LISTS +		Inquires the identifiers of the 3D graphics lists currently shown in the display.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]*	c		e	f	g	a		i		j	k	l	m	n	b
	3D graphics list identifiers	Specifies the identifiers of the 3D graphics lists currently shown in the display.	c		e	f	g	a		i		j	k	l	m	n	b

M_SIZE_X +	Inquires the horizontal resolution of the window. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the horizontal resolution, in pixels. INFO The default value is 800.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Inquires the vertical resolution of the window. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 1	Specifies the vertical resolution, in pixels. INFO The default value is 600.	c		e	f	g	a		i		j	k	l	m	n	b
M_TITLE +	Inquires the 3D display's title. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *array of type AIL_TEXT_CHAR [optionally, in C++: AIL_STRING]*	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*Title*") 1	Specifies the title of the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSPARENCY_SORT_MODE +	Inquires which depth sorting method to use when semi-opaque (semi-transparent) 3D graphics are shown in the 3D display. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEPTH_PEELING	Specifies to use depth peeling. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_FAST	Specifies to use simple depth sorting. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_UPDATE +	Inquires whether Aurora Imaging Library will update the 3D display. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies not to update the 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to update the 3D display. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_UPDATE_GRAPHIC_LIST +		Inquires whether Aurora Imaging Library should update the 3D display when modifications are made to the display's associated 2D graphics list. To associate a 2D graphics list to the 3D display, use M_ASSOCIATED_GRAPHIC_LIST_ID. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the 3D display is not updated when the associated 2D graphics list is modified.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the 3D display is immediately updated when the associated 2D graphics list is modified (refresh not required). This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_BOX_NODE +		Inquires to which 3D graphic the view will be reset when you use M3ddispSetView() with M_AUTO. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the label of a 3D graphic in the 3D display's 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_INITIAL_POSITION_X +		Inquires the initial left-most X-coordinate of the window. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the X-coordinate, in pixels. INFO The default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_INITIAL_POSITION_Y +		Inquires the initial top-most Y-coordinate of the window. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the Y-coordinate, in pixels. INFO The default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_MAXBUTTON +		Inquires whether the window's maximize button is visible. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the button is not visible.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the button is visible. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_MINBUTTON +		Inquires whether the window's minimize button is visible. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the button is not visible.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the button is visible. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_MOVE +		Inquires whether window movement is allowed. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to allow window movement.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to allow window movement. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_OVERLAP +		Inquires whether the window can be overlapped by another. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies not to allow the window to be overlapped by another (keep window on top).	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to allow the window to be overlapped by another. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_RESIZE +		Inquires whether window resizing is allowed. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to allow window resizing.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to allow window resizing. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_SHOW +		Inquires whether the window should be shown or hidden. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the window should be hidden.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the window should be shown. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_SYSBUTTON +		Inquires whether the window's system buttons are visible. The system buttons refer to the collection of the window's minimize, maximize, and close buttons. To individually set whether the minimize and maximize buttons are visible, change the M_WINDOW_MAXBUTTON or M_WINDOW_MINBUTTON control type settings. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the buttons are not visible.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the buttons are visible. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_WINDOW_TITLE_BAR +		Inquires whether the window's title bar is visible. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the title bar is not visible.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the title bar is visible. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If the value was passed in a variable, it won't be returned enclosed in AIL_TEXT().

The following InquireType and corresponding UserVarPtr parameter settings are available for picking contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For picking contexts																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_BLOCKABLE +		Inquires whether 3D graphics are blocked from being picked if the closest 3D graphic at a given pixel is one that has been filtered out. Note, this setting is only available for point picking contexts. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies to ignore filtered out 3D graphics entirely. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to acknowledge filtered out 3D graphics. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_FACES_AND_LINES +		Inquires whether faces and lines can be picked. This inquire type only applies to meshed point cloud graphics, arc graphics, line graphics, lines graphics, filled graphics, and graphics with M_APPEARANCE set to M_SOLID or M_WIREFRAME. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that faces and lines are not pickable.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that faces and lines are pickable. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_TYPE +		Inquires which type of 3D graphic can be detected by the pick operation. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies any type of 3D graphic. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_ARC	Specifies an arc graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_ARC_FILL	Specifies a filled arc graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_AXIS	Specifies an axis graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_BOX	Specifies a box graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_CYLINDER	Specifies a cylinder graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_GRAPHIC_TYPE_DOTS	Specifies dots graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_GRID	Specifies a grid graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINE	Specifies a line graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_LINES	Specifies a lines graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_PLANE	Specifies a plane graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POINT_CLOUD	Specifies a point cloud graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_POLYGON	Specifies a polygon graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_RECT	Specifies a rectangle graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_SPHERE	Specifies a sphere graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_TYPE_TEXT	Specifies text graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL +		Inquires which label is required for a 3D graphic to be detected by the pick operation. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_LABEL	Specifies to not filter out 3D graphics according to their label. INFO This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 1	Specifies the label of the 3D graphic that is pickable.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY_THRESHOLD +		Inquires the minimum opacity required for a 3D graphic to be detected by the pick operation. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 100.0	Specifies the opacity threshold. INFO The default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS +		Inquires whether points can be picked. This inquire type only applies to point cloud graphics, dots graphics, and graphics with M_APPEARANCE set to M_POINTS. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that points are not pickable.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that points are pickable. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_RENDER_LAYER_MASK +		Inquires on which layers a 3D graphic can be detected by the pick operation. Graphics on higher levels are always detected (and drawn before) those on lower layers. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

	M_LAYER_MASK(AIL_INT64 Value)	Specifies to only detect 3D graphics on the specified layer.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Value Specifies the layer. The layer must be between 0 and 9. Note, you can combine multiple values using the bitwise operator, "|". For example, specify M_LAYER_MASK(0) | M_LAYER_MASK(4) | M_LAYER_MASK(5) to mask out all layers other than 0, 4, and 5.															
	M_LAYER_UNMASK(AIL_INT64 Value)	Specifies to detect 3D graphics on all but the specified layer.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Value Specifies the layer. The layer must be between 0 and 9. Note, you can combine multiple values using the bitwise operator, "&". For example, specify M_LAYER_UNMASK(0) & M_LAYER_UNMASK(4) & M_LAYER_UNMASK(5) to mask out layers 0, 4, and 5.															
	M_ANY	Specifies to detect 3D graphics on any layer. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b

	0*b*0000000000 <= *Value* <= 0*b*1111111111	Specifies a bit mask, where the status of bits from 0 to 9 indicate which layers to mask; only 3D graphics on the layers that correspond to bits with non-zero values will be detected. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SELECTION_RADIUS +		Inquires the approximate radius around the specified position for the pick operation to search. Note, this setting is only available for point picking contexts. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1.0	Specifies the selection radius, in pixels. The default value is 8.0.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_TITLE.

You can add the following value to the above-mentioned value to get the string's length.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For getting the string size																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_STRING_SIZE		Retrieves the length of the string, including the terminating null character ("\0").															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring keyboard and mouse controls
- For 3D displays
- For picking contexts

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to determine whether an inquire type is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP

Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring whether an inquire type is supported																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_HAS_DEFAULT		Inquires whether the specified inquire type has a default value.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUPPORTED		Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_ALTERNATE_SPEED_FACTOR; M_BACKGROUND_COLOR; M_BACKGROUND_COLOR_GRADIENT; M_BACKGROUND_MODE; M_ROTATION_SPEED.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP
Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_DEFAULT		Inquires the default value of the specified inquire type.															
	UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT		Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispLut

Synopsis

Set a LUT to use to color the points of point cloud graphics in a 3D display.

Syntax

```
void M3ddispLut(
    AIL_ID Display3dId,               //out
    AIL_INT64 DisplayPointCloudLabel, //in
    AIL_ID LutBufId,                  //in
    AIL_INT64 Operation               //in
)
```

Description

This function sets the LUT buffer to use for point cloud graphics selected to the specified 3D display. **M3ddispLut**() maps the third band of the source point cloud's range component through the specified LUT.

You can specify to set a LUT for all point cloud graphics selected to the display, or for a specific point cloud graphic.

> If you set the LUT for all point cloud graphics (M_ALL or M_DEFAULT), subsequent point cloud graphics selected to the same 3D display (using M3ddispSelect()) will use the specified LUT.

For more control when applying a LUT buffer (for example, to apply the LUT to a different band or a different component altogether), use M3dgraCopy() instead of **M3ddispLut**(). Specifically, call M3dgraCopy() with M_COLOR_LUT to associate the LUT with the internal 3D graphics list; then, use M3dgraControl() with M_COLOR_USE_LUT set to M_TRUE, and M_COLOR_COMPONENT and M_COLOR_COMPONENT_BAND set to the required component and band, respectively.

To cease using a LUT with point clouds shown in the 3D display, call **M3ddispLut**() with LutBufId set to M_DEFAULT.

For more information, see the Lookup tables in general section of Chapter 24: Lookup tables and the Applying LUTs subsection of the Color and display settings for 3D data section of Chapter 43: 3D Display and graphics.

Parameters

Display3dId

Specifies the identifier of the 3D display for which to set the LUT buffer.

DisplayPointCloudLabel

Specifies the label of the point cloud graphic to which to apply the LUT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the label																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to apply the LUT to all point cloud graphics currently and subsequently selected to the 3D display. This is also applied to point cloud graphics that were added using M3dgraAdd(). Note that this setting changes the default settings of the internal 3D graphics list (M3dgraControl() with M_DEFAULT_SETTINGS and M_COLOR_USE_LUT, M_COLOR_COMPONENT, M_COLOR_COMPONENT_BAND).	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the label of the point cloud graphic whose LUT to set. This is typically the return value of a call to M3ddispSelect().	c		e	f	g	a		i		j	k	l	m	n	b

LutBufId

Specifies the identifier of a LUT buffer. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the identifier of the previously allocated LUT buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies not to use a LUT buffer. All LUT settings are restored to their defaults. The point cloud graphic is colored based on its source container's reflectance or intensity component (M3dgraControl() with M_COLOR_COMPONENT set to M_AUTO_COLOR).	c		e	f	g	a		i		j	k	l	m	n	b
M_COLORMAP_GRAYSCALE +	Specifies to use a predefined LUT buffer with a grayscale colormap as the LUT of a point cloud graphic in a 3D graphics list. The predefined LUT buffer has 3 bands and has a size of 65536. The colormap transitions from black to white as the indices increase. Note that this colormap is effectively the same as no colormap, except that you can apply the M_FLIP combination value to invert the grayscale values.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLORMAP_HOT +	Specifies to use a predefined LUT buffer with a hot colormap as the LUT of a point cloud graphic in a 3D graphics list. The predefined LUT buffer has 3 bands and has a size of 65536. The colormap transitions from black, to red, to yellow, and then, to white along the RGB cube as the indices increase. This colormap can be useful for displaying infrared images.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLORMAP_HUE +	Specifies to use a predefined LUT buffer with a hue colormap as the LUT of a point cloud graphic in a 3D graphics list. The predefined LUT buffer has 3 bands and has a size of 65536. The colormap is cyclic and transitions from red, to yellow, to green, to cyan, to blue, to magenta, and then, to red along the edge of the hue circle as the indices increase. This colormap is useful for displaying the hue component of an HSL image.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLORMAP_JET +	Specifies to use a predefined LUT buffer with a jet colormap as the LUT of a point cloud graphic in a 3D graphics list. The predefined LUT buffer has 3 bands and has a size of 65536. The colormap transitions from dark blue, to blue, to cyan, to yellow, to red, and then, to dark red along the RGB cube as the indices increase. This colormap can be useful for displaying 3D elevation maps and thermal imaging.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLORMAP_SPECTRUM +	Specifies to use a predefined LUT buffer with a spectrum colormap as the LUT of a point cloud graphic in a 3D graphics list. The predefined LUT buffer has 3 bands and has a size of 65536. The colormap transitions from red, to yellow, to green, to blue, and then, to violet along the RGB cube as the indices increase. This colormap is a representation of the visual spectrum according to wavelengths. It can also be useful for displaying 3D elevation maps and thermal imaging.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLORMAP_TURBO +	Specifies to use a predefined LUT buffer with a turbo colormap as the LUT of a point cloud graphic in a 3D graphics list. The predefined LUT buffer has 3 bands and has a size of 65536. The colormap transitions from dark blue, to blue, to cyan, to yellow, to red, and then, to dark red along the RGB cube as the indices increase. This colormap is similar to M_COLORMAP_JET, but with smoother transitions.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLORMAP_WHEEL +	Specifies to use a predefined LUT buffer with a colorwheel colormap as the LUT of a point cloud graphic in a 3D graphics list. The predefined LUT buffer has 3 bands and has a size of 65536. The colormap is cyclic and transitions from blue, to magenta, to yellow, to green, and then, to blue again as the indices increase. This colormap is designed to be perceptually uniform (a given change in data is designed to be visually perceived as an equal change in color). This colormap can be useful for displaying 3D elevation maps.	c		e	f	g	a		i		j	k	l	m	n	b
LUT buffer ID +	Specifies the identifier of the LUT buffer to use as the LUT of a point cloud graphic in a 3D graphics list. The LUT buffer must have been previously allocated using MbufAlloc1d() or MbufAllocColor() with M_LUT. Only 8-bit unsigned LUT buffers are supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For the identifier of the previously allocated LUT buffer.

You can add the following value to the above-mentioned values to specify that the LUT should be flipped.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying to flip the LUT																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FLIP	Specifies to flip the LUT's values (inverse LUT).	c		e	f	g	a		i		j	k	l	m	n	b

Operation

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispPick

Synopsis

Pick a 3D graphic at a given pixel in a 3D display.

Syntax

```
void M3ddispPick(
    AIL_ID Context3ddispId, //in
    AIL_ID Disp3dId,        //in
    AIL_ID Result3ddispId,  //out
    AIL_INT64 PixelX,       //in
    AIL_INT64 PixelY,       //in
    AIL_INT64 ControlFlag   //in
)
```

Description

This function identifies which 3D graphic (if any) is at a specific position in the 3D display and writes the results in the specified result buffer. You can retrieve results using M3ddispGetResult(). Aurora Imaging Library identifies the 3D graphic by casting a ray from the 3D display's viewpoint towards the given pixel and detecting with which 3D graphic that the ray intersects. Note that when there is more than one 3D graphic at a given pixel, the detected 3D graphic is the one that appears in front. If the 3D graphics are on the same layer, the 3D graphic closest to the viewpoint is detected. If the 3D graphics are on different layers, the 3D graphic on the highest layer is detected, regardless of its position in the world.

Use M3ddispControl() to configure the settings of the picking context before calling this function. You can filter out 3D graphics that you don't want to pick. For example, you can use M_GRAPHIC_TYPE to only detect 3D graphics of a certain type. If a 3D graphic of the specified type is behind a filtered out 3D graphic (either on a lower layer or further away on the same layer), the ray will pass through the filtered out 3D graphic and **M3ddispPick**() will detect the specified graphic. You can set M_BLOCKABLE to M_ENABLE to allow the filtered out 3D graphic to block the ray.

This function is useful for identifying which 3D graphic was clicked on. To do so, you can set PixelX and PixelY to the X- and Y-positions of your cursor (M3ddispGetHookInfo() with M_MOUSE_POSITION_X and M_MOUSE_POSITION_Y) when an M_MOUSE_LEFT_BUTTON_DOWN event is triggered.

Parameters

Context3ddispId

Specifies the identifier of the picking context. The picking context must have been previously allocated using M3ddispAlloc() with M_PICKING_CONTEXT.

Disp3dId

Specifies the identifier of the 3D display. The 3D display must have been previously allocated using M3ddispAlloc().

Result3ddispId

Specifies the identifier of the picking result buffer in which to write the results. The picking result buffer must have been previously allocated using M3ddispAllocResult() with M_PICKING_RESULT.

PixelX

Specifies the X-coordinate of the pixel through which to cast the ray. If the specified pixel coordinate is greater than the 3D display's current M_SIZE_X, an error is generated.

PixelY

Specifies the Y-coordinate of the pixel through which to cast the ray. If the specified pixel coordinate is greater than the 3D display's current M_SIZE_Y, an error is generated.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispPickRect

Synopsis

Pick 3D graphics within a given pixel range defined by a rectangular region in a 3D display.

Syntax

```
void M3ddispPickRect(
    AIL_ID Context3ddispId, //in
    AIL_ID Disp3dId,        //in
    AIL_ID Result3ddispId,  //out
    AIL_INT64 PixelX1,      //in
    AIL_INT64 PixelY1,      //in
    AIL_INT64 PixelX2,      //in
    AIL_INT64 PixelY2,      //in
    AIL_INT64 ControlFlag   //in
)
```

Description

This function identifies which 3D graphics (if any) are in a specific rectangular region in the 3D display and writes the results in the specified result buffer. You can retrieve results using M3ddispGetResult(). Aurora Imaging Library identifies the 3D graphics by casting a frustum from the 3D display's viewpoint towards the given region and detecting the 3D graphic(s) that intersect the frustum (either completely or partially). Note that 3D graphics occluded by other 3D graphics are still picked, provided that they are within the frustum.

You can set a region that encapsulates the 3D display in its entirety, a portion of it, or none of it at all. Any 3D graphics intersecting the resulting frustum will be picked, including those outside the visible portion of the 3D display.

Use M3ddispControl() to configure the settings of the picking context before calling this function. You can filter out 3D graphics that you don't want to pick. For example, you can use M_GRAPHIC_TYPE to only detect 3D graphics of a certain type. If 3D graphics of the specified type are behind filtered out 3D graphics (either on a lower layer or further away on the same layer), the frustum will pass through the filtered out graphics and **M3ddispPickRect**() will detect the specified graphics.

To define the rectangular region, use **M3ddispPickRect**() to specify the X- and Y-coordinates of 2 separate pixels. These will form opposite corners of the rectangle from which to pick. You can select the required pixels interactively. To do so, set the 4 pixel coordinate parameters (for example, PixelX1) to the X- and Y-positions of your cursor for 2 consecutive mouse click events. Use M3ddispGetHookInfo() with M_MOUSE_POSITION_X and M_MOUSE_POSITION_Y to get the position of the mouse when an M_MOUSE_LEFT_BUTTON_DOWN event is triggered.

Parameters

Context3ddispId

Specifies the identifier of the picking area context. The picking area context must have been previously allocated using M3ddispAlloc() with M_PICKING_AREA_CONTEXT.

Disp3dId

Specifies the identifier of the 3D display. The 3D display must have been previously allocated using M3ddispAlloc().

Result3ddispId

Specifies the identifier of the picking area result buffer in which to write the results. The picking area result buffer must have been previously allocated using M3ddispAllocResult() with M_PICKING_AREA_RESULT.

PixelX1

Specifies the first X-coordinate of the region from which to generate the picking frustum. If the specified pixel coordinate is outside the bounds of the display (less than 0 or greater than M_SIZE_X), a warning is generated. 3D graphics outside of the bounds of the display are still detected and picked.

PixelY1

Specifies the first Y-coordinate of the region from which to generate the picking frustum. If the specified pixel coordinate is outside of the bounds of the display (less than 0 or greater than M_SIZE_Y), a warning is generated. 3D graphics outside of the bounds of the display are still detected and picked.

PixelX2

Specifies the second X-coordinate of the region from which to generate the picking frustum. If the specified pixel coordinate is outside of the bounds of the display (less than 0 or greater than M_SIZE_X), a warning is generated. 3D graphics outside of the bounds of the display are still detected and picked.

PixelY2

Specifies the second Y-coordinate of the region from which to generate the picking frustum. If the specified pixel coordinate is outside of the bounds of the display (less than 0 or greater than M_SIZE_Y), a warning is generated. 3D graphics outside of the bounds of the display are still detected and picked.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispSelect

Synopsis

Add a point cloud graphic, generated from a point cloud container or depth map image buffer/container, or add a graphics list to the specified 3D display. The function can also remove them from the display. Furthermore, the function can open or close the window associated with the 3D display.

Syntax

```
AIL_INT64 M3ddispSelect(
    AIL_ID Disp3dId,      //out
    AIL_ID AilObjectId,   //in
    AIL_INT64 Option,     //in
    AIL_INT64 ControlFlag //in
)
```

Description

This function can add a point cloud graphic, generated from a point cloud container or depth map image buffer/container to the 3D display's internal graphics list, or can add a 3D graphics list to the specified Aurora Imaging Library 3D display. This function also optionally opens or closes a window to show the 3D display, or removes a specified point cloud graphic or graphics list previously added to the 3D display.

The specified container or image buffer is linked to the point cloud graphic, which is added to the 3D display's internal 3D graphics list.

> Any subsequent changes made to the image buffer or components of the container will be reflected in the 3D display. The point cloud container or depth map image buffer/container must be allocated for display (M_DISP); otherwise, no point cloud graphic will be shown. You can determine whether a container is currently 3D-displayable using MbufInquireContainer() with M_3D_DISPLAYABLE.

By default, this function removes any point cloud graphics already selected to the 3D display. To show multiple point cloud graphics in the 3D display simultaneously, you can either specify the M_ADD option, or use M3dgraAdd(). To remove only a single point cloud graphic from the 3D display, specify the M_REMOVE option, or use M3dgraRemove(). Note that you can use M3dgraAdd() with M_NO_LINK to display an unlinked point cloud graphic; in this case, the 3D display is not updated dynamically.

If you want to annotate the 3D display, inquire either the identifier of its internal 3D graphics list or the identifier of a 3D graphics list previously added to the display, using M3ddispInquire() with M_3D_GRAPHIC_LIST_ID. Then, add graphic objects to the 3D graphics list using the functions in the M3dgra...() module.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call functions in this module concurrently from multiple threads on the same Aurora Imaging Library 3D display (Disp3dId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data. Point cloud containers and depth map image buffers/containers are **not thread-safe** and must only be accessed by one thread at a time. You must ensure the source point cloud container or depth map image buffer/container is not modified from another thread while the add is being performed. Note that this does not impact subsequent modifications to the point cloud container or depth map image buffer/container as long as they are modified from one thread at a time.

Parameters

Disp3dId

Specifies the identifier of the 3D display, previously allocated using M3ddispAlloc().

AilObjectId

Specifies the Aurora Imaging Library identifier of the 3D-displayable container, fully corrected depth map image buffer, or graphics list to add to the display.

> Note that the container, image buffer, or 3D graphics list and the 3D display should be allocated on the same system.

See the Parameter associations section for possible values that can be specified.

Option

Specifies the action to perform on the specified 3D display.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The table below lists possible values for the **AilObjectId** and **Option** parameters.

- For specifying the container, image buffer, or 3D graphics list and what operation to perform

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the container, image buffer, or 3D graphics list and what operation to perform																	
AilObjectId		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	Option																
M_NULL		Specifies no container, image buffer, or 3D graphics list.															
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSE	Specifies to close the window. > This setting is not available for M_WEB displays.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPEN	Specifies to open the window. > This setting is not available for M_WEB displays.	c		e	f	g	a		i		j	k	l	m	n	b

	M_REMOVE	Specifies to remove all point cloud graphics from the 3D display's internal 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to remove all point cloud graphics from the 3D display's internal 3D graphics list and close the window.	c		e	f	g	a		i		j	k	l	m	n	b
3D-displayable container identifier		Specifies the Aurora Imaging Library identifier of the 3D-displayable point cloud or depth map container, for adding or removing its linked point cloud graphic to/from the specified 3D display's internal 3D graphics list. You can inquire whether a container is 3D-displayable using MbufInquireContainer() with M_3D_DISPLAYABLE. If the container is only 3D-displayable with compensation, Aurora Imaging Library will perform an internal conversion each time the container is modified. For maximum performance, you should convert the container to a format that is natively 3D-displayable (using MbufConvert3d() with a destination container that has the attribute M_DISP). M3dimFix() with M_RANGE_FINITE + M_NORMALS_FINITE + M_MESH_VALID_POINTS is internally performed for point cloud graphics. Just like automatic compensation, this does not modify the source point cloud.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ADD	Specifies to add the point cloud graphic, linked to the specified container or depth map image buffer, to the 3D display's internal 3D graphics list. If the specified container or image buffer is already linked to a point cloud graphic in the 3D graphics list, no error is generated. The function will return the label of the existing point cloud graphic instead. If, however, the existing point cloud graphic is unlinked (that is, it was added using M3dgraAdd() with M_NO_LINK), using this option adds a duplicate point cloud to the 3D graphics list. This does not affect the open/close status of the display. Note that to add multiple linked point cloud graphics, use M3dgraAdd().	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE	Specifies to remove, from the display's internal graphics list, all point cloud graphic labels corresponding to the specified container. An error is generated if the specified container is not linked to a point cloud graphic in the internal 3D graphics list. You can remove unlinked point cloud graphics from the display's internal graphics list using M3dgraRemove().	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to add the specified container's point cloud graphic to the 3D display's internal 3D graphics list, remove all other point cloud graphics from the internal 3D graphics list, and open the window. Note, an error will occur if a user-added graphics list has already been added to the display.	c		e	f	g	a		i		j	k	l	m	n	b
3D graphics list identifier		Specifies the Aurora Imaging Library identifier of the 3D graphics list to add to or remove from the specified 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ADD	Specifies to add the 3D graphics list to the display. This does not affect the open/closed status of the display.	c		e	f	g	a		i		j	k	l	m	n	b

	M_REMOVE	Specifies to remove the 3D graphics list from the display. Note, the default graphics list cannot be removed. If the specified 3D graphics list is not associated with the display, an error will occur.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to remove all 3D graphics lists (excluding the default graphics list) and add the specified graphics list to the display. Note, if the default graphics list is not empty, an error will occur.	c		e	f	g	a		i		j	k	l	m	n	b
Empty container identifier		Specifies the Aurora Imaging Library identifier of an empty container with the M_DISP attribute. This is useful for specifying a container into which you will subsequently grab 3D data.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ADD	Specifies to add the point cloud graphic, linked to the specified container or depth map image buffer, to the 3D display's internal 3D graphics list. If the specified container is already linked to a point cloud graphic in the 3D graphics list, no error is generated. The function will return the label of the existing point cloud graphic instead. If, however, the existing point cloud graphic is unlinked (that is, it was added using M3dgraAdd() with M_NO_LINK), using this option adds a duplicate point cloud to the 3D graphics list. Since the specified container doesn't initially contain a point cloud or depth map, a point cloud graphic without any points is added.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE	Specifies to remove the point cloud graphic, linked to the specified container, from the 3D display's internal 3D graphics list. An error is generated if the specified container is not linked to a point cloud graphic in the internal 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SELECT	Specifies to add the specified container's point cloud graphic to the 3D display's internal 3D graphics list, remove all other point cloud graphics from the internal 3D graphics list, and open the window. Since the specified container doesn't initially contain a point cloud or depth map, a point cloud graphic without any points is added.	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier		Specifies the Aurora Imaging Library identifier of an image buffer, with the M_DISP attribute, for adding or removing its linked point cloud graphic to/from the specified 3D display's internal 3D graphics list. This image buffer will be displayed as a point cloud. The image buffer must be a fully-corrected depth map (McalInquire() with M_DEPTH_MAP returns M_TRUE).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ADD	Specifies to add the point cloud graphic, linked to the specified depth map image buffer, to the 3D display's internal 3D graphics list. If the specified image buffer is already linked to a point cloud graphic in the 3D graphics list, no error is generated. The function will return the label of the existing point cloud graphic instead. If, however, the existing point cloud graphic is unlinked (that is, it was added using M3dgraAdd() with M_NO_LINK), using this option adds a duplicate point cloud to the 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE	Specifies to remove the point cloud graphic, linked to the specified depth map image buffer, from the 3D display's internal 3D graphics list. An error is generated if the specified image buffer is not linked to a point cloud graphic in the internal 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SELECT	Specifies to add the specified image buffer's point cloud graphic to the 3D display's internal 3D graphics list, remove all other point cloud graphics from the internal 3D graphics list, and open the window.	c		e	f	g	a		i		j	k	l	m	n	b	

Return value

Returns the label of the point cloud graphic added to the internal 3D graphics list of the 3D display.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispSelectWindow

Synopsis

Add a point cloud graphic, generated from a point cloud container or depth map image buffer/container, or add a graphics list to a 3D display, and associate the 3D display with a user-defined window. The function can also remove them from the 3D display.

Syntax

```
AIL_INT64 M3ddispSelectWindow(
    AIL_ID Disp3dId,                    //out
    AIL_ID AilObjectId,                 //in
    AIL_INT64 Option,                   //in
    AIL_INT64 ControlFlag,              //in
    AIL_WINDOW_HANDLE ClientWindowHandle //in
)
```

Description

This function can add a point cloud graphic, generated from a point cloud container or depth map image buffer/container, to a 3D display's internal graphics list or can add a graphics list to the 3D display. It can associate and show the 3D display in a user-defined window. This function can also remove a specified point cloud graphic or graphics list previously added to the 3D display.

> This function never affects the user-defined window's open or closed status; the function only affects the 3D display's association with it.

The specified container or image buffer is linked to the point cloud graphic, which is added to the 3D display's internal 3D graphics list.

> Any subsequent changes made to the image buffer or components of the container will be shown in the 3D display. The point cloud container or depth map image buffer/container must be allocated for display (M_DISP); otherwise, no point cloud graphic will be shown. You can determine whether a container is currently 3D-displayable using MbufInquireContainer() with M_3D_DISPLAYABLE.

By default, this function removes any point cloud graphics already selected to the 3D display. To show multiple point cloud graphics in the 3D display simultaneously, you can either specify the M_ADD option, or use M3dgraAdd(). To remove only a single container or image buffer from the 3D display, specify the M_REMOVE option, or use M3dgraRemove(). Note that you can use M3dgraAdd() with M_NO_LINK to display an unlinked point cloud graphic; in this case, the 3D display is not updated dynamically.

If you want to annotate the 3D display, inquire either the identifier of its internal 3D graphics list or the identifier of a 3D graphics list previously added to the display, using M3ddispInquire() with M_3D_GRAPHIC_LIST_ID. Then, add graphic objects to the 3D graphics list using the functions in the M3dgra...() module.

> Unlike most other functions that modify an Aurora Imaging Library object, you can call functions in this module concurrently from multiple threads on the same Aurora Imaging Library 3D display (Disp3dId) without using an M_MUTEX object, as long as all the other parameters of the concurrent calls do not also share data. Point cloud containers and depth map image buffers/containers are **not thread-safe** and must only be accessed by one thread at a time. You must ensure the source point

> cloud container or depth map image buffer/container is not modified from another thread while the add is being performed. Note that this does not impact subsequent modifications to the point cloud container or depth map image buffer/container as long as they are modified from one thread at a time.

Parameters

Disp3dId

Specifies the identifier of the 3D display, previously allocated using M3ddispAlloc().

> This function is not supported for Aurora Imaging Web Library (M_WEB) 3D displays. To select a container or image buffer to an Aurora Imaging Web Library 3D display, use M3ddispSelect().

AilObjectId

Specifies the Aurora Imaging Library identifier of the 3D-displayable container, fully corrected depth map image buffer, or graphics list to add to the display.

> Note that the container, image buffer, or 3D graphics list and the 3D display should be allocated on the same system.

Option

Specifies the action to perform on the specified 3D display.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

See the Parameter associations section for possible values that can be specified.

ClientWindowHandle

Specifies the window handle of the user-defined window or child window. This window must have a window handle of type *HWND* or X11. For example, the Windows API functions can create a window with an *HWND* handle, and third-party functions (like Qt) can create an X11 window handle. If this parameter is set to zero, this function behaves like M3ddispSelect().

Set this parameter to **M_NULL** if AilObjectId is set to **M_NULL**.

Parameter associations

The table below lists possible values for the **Option** and **ControlFlag** parameters.

- For specifying the container, image buffer, or 3D graphics list and what operation to perform

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the container, image buffer, or 3D graphics list and what operation to perform																	
Option		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlFlag																

M_NULL		Specifies no container, image buffer, or 3D graphics list.														
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSE	Specifies to stop showing the 3D display and to disassociate it from the user-defined window. > This setting is not available for M_WEB displays.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPEN	Specifies to associate the 3D display with the user-defined window and to show the 3D display when the window is open. > This setting is not available for M_WEB displays.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE	Specifies to remove all point cloud graphics from the 3D display's internal 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to remove all point cloud graphics from the 3D display's internal 3D graphics list and to disassociate the 3D display from the user-defined window.	c		e	f	g	a		i		j	k	l	m	n	b

3D-displayable container identifier		Specifies the Aurora Imaging Library identifier of the 3D-displayable point cloud or depth map container, for adding or removing its linked point cloud graphic to/from the specified 3D display's internal 3D graphics list. You can inquire whether a container is 3D-displayable using MbufInquireContainer() with M_3D_DISPLAYABLE. If the container is only 3D-displayable with compensation, Aurora Imaging Library will perform an internal conversion each time the container is modified. For maximum performance, you should convert the container to a format that is natively 3D-displayable (using MbufConvert3d() with a destination container that has the attribute M_DISP). M3dimFix() with M_RANGE_FINITE + M_NORMALS_FINITE + M_MESH_VALID_POINTS is internally performed for point cloud graphics. Just like automatic compensation, this does not modify the source point cloud.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ADD	Specifies to add the point cloud graphic, linked to the specified container or depth map image buffer, to the 3D display's internal 3D graphics list. If the specified container or image buffer is already linked to a point cloud graphic in the 3D graphics list, no error is generated. The function will return the label of the existing point cloud graphic instead. If, however, the existing point cloud graphic is unlinked (that is, it was added using M3dgraAdd() with M_NO_LINK), using this option adds a duplicate point cloud to the 3D graphics list. This does not affect the association of the 3D display with the user-defined window. Note that to add multiple linked point cloud graphics, use M3dgraAdd().	c		e	f	g	a		i		j	k	l	m	n	b

	M_REMOVE	Specifies to remove, from the display's internal graphics list, all point cloud graphic labels corresponding to the specified container. An error is generated if the specified container is not linked to a point cloud graphic in the internal 3D graphics list. You can remove unlinked point cloud graphics from the display's internal graphics list using M3dgraRemove().	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to add the specified container's point cloud graphic to the 3D display's internal 3D graphics list, remove all other point cloud graphics from the internal 3D graphics list, and associate the 3D display with the user-defined window. Note, an error will occur if a user-added graphics list has already been added to the display.	c		e	f	g	a		i		j	k	l	m	n	b
3D graphics list identifier		Specifies the Aurora Imaging Library identifier of the 3D graphics list to add to or remove from the specified 3D display.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ADD	Specifies to add the 3D graphics list to the display. This does not affect the association of the 3D display with the user-defined window.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE	Specifies to remove the 3D graphics list from the display. Note, the default graphics list cannot be removed. If the specified 3D graphics list is not associated with the display, an error will occur.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to remove all 3D graphics lists (excluding the default graphics list) and add the specified graphics list to the display. This does not affect the association of the 3D display with the user-defined window. Note, if the default graphics list is not empty, an error will occur.	c		e	f	g	a		i		j	k	l	m	n	b

Empty container identifier		Specifies the Aurora Imaging Library identifier of an empty container with the M_DISP attribute. This is useful for specifying a container into which you will subsequently grab 3D data.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ADD	Specifies to add the point cloud graphic, linked to the specified container or depth map image buffer, to the 3D display's internal 3D graphics list. If the specified container is already linked to a point cloud graphic in the 3D graphics list, no error is generated. The function will return the label of the existing point cloud graphic instead. If, however, the existing point cloud graphic is unlinked (that is, it was added using M3dgraAdd() with M_NO_LINK), using this option adds a duplicate point cloud to the 3D graphics list. Since the specified container doesn't initially contain a point cloud or depth map, a point cloud graphic without any points is added. This does not affect the association of the 3D display with the user-defined window.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE	Specifies to remove the point cloud graphic, linked to the specified container, from the 3D display's internal 3D graphics list. An error is generated if the specified container is not linked to a point cloud graphic in the internal 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to add the specified container's point cloud graphic to the 3D display's internal 3D graphics list, remove all other point cloud graphics from the internal 3D graphics list, and associate the 3D display with the user-defined window. Since the specified container doesn't initially contain a point cloud or depth map, a point cloud graphic without any points is added.	c		e	f	g	a		i		j	k	l	m	n	b

Image buffer identifier		Specifies the Aurora Imaging Library identifier of an image buffer, with the M_DISP attribute, for adding or removing its linked point cloud graphic to/from the specified 3D display's internal 3D graphics list. This image buffer will be displayed as a point cloud. The image buffer must be a fully-corrected depth map (McalInquire() with M_DEPTH_MAP returns M_TRUE).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SELECT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ADD	Specifies to add the point cloud graphic, linked to the specified depth map image buffer, to the 3D display's internal 3D graphics list. If the specified image buffer is already linked to a point cloud graphic in the 3D graphics list, no error is generated. The function will return the label of the existing point cloud graphic instead. If, however, the existing point cloud graphic is unlinked (that is, it was added using M3dgraAdd() with M_NO_LINK), using this option adds a duplicate point cloud to the 3D graphics list. This does not affect the association of the 3D display with the user-defined window.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REMOVE	Specifies to remove the point cloud graphic, linked to the specified depth map image buffer, from the 3D display's internal 3D graphics list. An error is generated if the specified image buffer is not linked to a point cloud graphic in the internal 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SELECT	Specifies to add the specified image buffer's point cloud graphic to the 3D display's internal 3D graphics list, remove all other point cloud graphics from the internal 3D graphics list, and associate the 3D display with the user-defined window.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

Returns the label of the point cloud graphic added to the 3D graphics list of the 3D display.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispSetHookInfo

Synopsis

Set information about an event that caused a hook-handler function to be called.

Syntax

```
void M3ddispSetHookInfo(
    AIL_ID EventId,      //out
    AIL_INT64 InfoType,  //in
    AIL_DOUBLE InfoValue //in
)
```

Description

This function allows you to set information about the event that caused the hook-handler function to be called. **M3ddispSetHookInfo**() must only be called within the scope of a 3d display object hook-handler function (see M3ddispHookFunction()).

Parameters

EventId

Specifies the display event identifier received by the hook-handler function. See M3ddispHookFunction() for more information. The event will be of the type passed to the HookType parameter of the hook-handler function.

InfoType

Specifies the type of information to set.

See the Parameter associations section for possible values that can be specified.

InfoValue

Specifies the required value for the type of information to set.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The table below lists possible values for the **InfoType** and **InfoValue** parameters.

- For setting whether the display will process any mouse or keyboard input events

To set information about the event that caused the hook-handler function to be called, the InfoType and corresponding InfoValue parameter settings can be set to one of the following values. These settings are only available if the hook-handler function was called due to mouse or keyboard input event (M_KEY_DOWN, M_KEY_UP, M_KEY_CHAR, M_MOUSE_LEFT_BUTTON_DOWN, M_MOUSE_LEFT_BUTTON_UP, M_MOUSE_LEFT_DOUBLE_CLICK, M_MOUSE_RIGHT_BUTTON_DOWN, M_MOUSE_RIGHT_BUTTON_UP M_MOUSE_MIDDLE_BUTTON_DOWN M_MOUSE_MIDDLE_BUTTON_UP, M_MOUSE_WHEEL, M_MOUSE_MOVE).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For setting whether the display will process any mouse or keyboard input events																		
InfoType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	InfoValue																
M_CONSUMED		Sets whether this callback will prevent the display from processes a mouse or keyboard input event after the hook-handler function has returned.															
	M_DEFAULT	Same as M_FALSE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the display will process the mouse or keyboard input event after the hook-handler function has returned. The callback will not consume the mouse or keyboard input event.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_CLEAR	Specifies that for mouse button down events (M_MOUSE_LEFT_BUTTON_DOWN, M_MOUSE_RIGHT_BUTTON_DOWN, M_MOUSE_MIDDLE_BUTTON_DOWN), the display's view can still be moved by moving the mouse after the mouse button down event. However, the initial click will not trigger the rotation indicator. Similar to M_TRUE but after moving the mouse the rotation indicator will also trigger. For example, by clicking on a point, you can consume the input event and display the coordinates of the pixel, and by moving the mouse you can simultaneously trigger the rotation indicator and move the view on the display. > Note that this only applies when M_ROTATION_INDICATOR is set M_ENABLE_ON_MOUSE_CLICK.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the display will not process the mouse or keyboard input event after the hook-handler function has returned. The callback will consume the mouse or keyboard input event.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.

M3ddispSetView

Synopsis

Sets the display's view.

Syntax

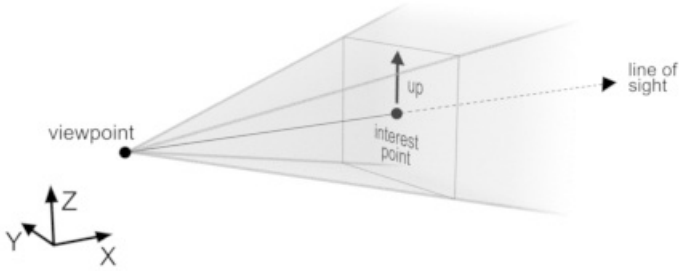
```
void M3ddispSetView(
    AIL_ID Disp3dId,      //out
    AIL_INT64 Mode,       //in
    AIL_DOUBLE Param1,    //in
    AIL_DOUBLE Param2,    //in
    AIL_DOUBLE Param3,    //in
    AIL_INT64 ControlFlag //in
)
```

Description

This function changes the position and orientation of the view of the specified 3D display.

> Note that, by default, you can also change the position of orientation of the view interactively using the keyboard and mouse. You can enable or disable the keyboard and mouse controls using M3ddispControl() with M_KEYBOARD_USE and M_MOUSE_USE.

This function changes the view of the 3D display by modifying the viewpoint, interest point, and/or up vector, as shown in the image below:



You can set the position of the viewpoint using M_VIEWPOINT, the position of the interest point using M_INTEREST_POINT and the direction of the up vector using M_UP_VECTOR. All other modes are alternatives to adjusting these aspects of the view directly. For example, setting the M_ROLL setting of the view is an alternate method to change the M_UP_VECTOR setting.

Most modes are specified in absolute terms (for example, using M_AZIMUTH with the value 90 sets the azimuth to 90, regardless of the current value). In some cases, you can instead specify to apply the operation relative to the current state of the view using M_COMPOSE_WITH_CURRENT. For example, if you use M_AZIMUTH with M_COMPOSE_WITH_CURRENT and the value 90, and the current azimuth is 30, the new azimuth is 120.

Modes that can be performed by moving either the viewpoint or the interest point will always move the viewpoint, unless you use the combination value M_MOVE_INTEREST_POINT. For example, if you use M_AZIMUTH with the value 45, by default the viewpoint is moved to the position that is rotated 45 degrees around the interest point (relative to the plane that intersects the interest point and is parallel to the ZY-plane), effectively circling the view around the interest point. If you specify

M_MOVE_INTEREST_POINT, instead the interest point is moved to the position in which the viewpoint is rotated 45 degrees around the interest point, effectively panning the view. Both modes result in the same orientation (the viewpoint is rotated 45 degrees around the interest point); however, in the first case the viewpoint has a new position, and in the second case the interest point has a new position.

Optionally, you can specify not to refresh the rendered image shown in the 3D display by setting the ControlFlag parameter to M_NO_REFRESH. This ControlFlag setting only guarantees that this operation will not trigger a refresh of the rendered image. It is possible for other events in the 3D display to trigger a refresh, including events which occur before or during the call to **M3ddispSetView**(). If you need to guarantee that there is no refresh between a sequence of operations, use M3ddispControl() with M_UPDATE to disable and enable rendering.

Parameters

Disp3dId

Specifies the identifier of the 3D display.

Mode

Specifies the mode used for setting the view.

See the Parameter associations section for possible values that can be specified.

Param1

Specifies the first parameter used to set the view.

See the Parameter associations section for possible values that can be specified.

Param2

Specifies the second parameter used to set the view.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Param3

Specifies the third parameter used to set the view.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Specifies if the 3D display should be refreshed.

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For specifying if the display should be updated																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies to trigger a refresh of the 3D display after the operation is complete.	c		e	f	g	a		i		j	k	l	m	n	b

M_NO_HOOK	Prevents generating an M_VIEW_CHANGE event in response to the call to **M3ddispSetView**(). If you want to prevent more hooks from being generated when calling **M3ddispSetView**() with M_VIEW_CHANGE for another 3D display (to which you are hooked).	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_REFRESH	Specifies not to trigger a refresh of the 3D display after the operation is complete. If you need to guarantee that there is no refresh between a sequence of operations, use M3ddispControl() with M_UPDATE to disable and enable rendering.	c		e	f	g	a		i		j	k	l	m	n	b

Parameter associations

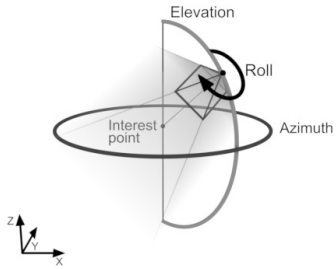
The table below lists possible values for the **Mode**, **Param1**, **Param2**, and **Param3** parameters.

- For setting the view

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

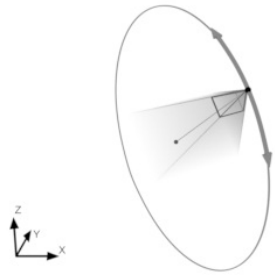
For setting the view																	
Mode		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	Param1																
	Param2																
	Param3																
M_DEFAULT		Same as M_AUTO.															
M_AUTO		Sets the viewpoint and interest point position so that a specific 3D graphic, and all of its descendants, are completely within the view. This is referred to as focusing the view on the 3D graphic. By default, the view is focused on the root node (this guarantees that everything in the 3D display's 3D graphics list is completely within the view). Optionally, you can also specify to set the orientation of the view to a preset. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Specifies the orientation of the view.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies to preserve the current orientation of the view.	c		e	f	g	a		i		j	k	l	m	n	b

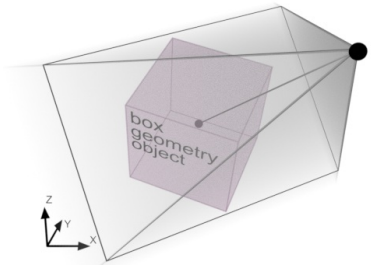
		M_BOTTOM_TILTED	Specifies to set the orientation of the view to an interpolation between the bottom, front, and left views.	c		e	f	g	a		i		j	k	l	m	n	b
		M_BOTTOM_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly below the interest point (parallel to the Z-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_FRONT_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly in front of the interest point (parallel to the Y-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_LEFT_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly to the left of the interest point (parallel to the X-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_REAR_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly behind the interest point (parallel to the Y-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_RIGHT_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly to the right of the interest point (parallel to the X-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_TOP_TILTED	Specifies to set the orientation of the view to an interpolation between the top, front, and left views.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TOP_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly above the interest point (parallel to the Z-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Specifies to which 3D graphic to reset the view.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies to focus the view on the 3D graphic that you set using M3ddispControl() with M_VIEW_BOX_NODE. By default, this is set to 0 (the root node).	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the label of the 3D graphic (in the 3D display's 3D graphics list) on which to focus the view.	c		e	f	g	a		i		j	k	l	m	n	b
M_AZIM_ELEV_ROLL +		Sets the azimuth and elevation of the viewpoint relative to the interest point, and the roll of the view. These are the viewpoint position and up vector, described in a spherical coordinate system (additionally described by M_DISTANCE). Azimuth is the angle of the viewpoint relative to the interest point on the XY-plane. An azimuth of 0 corresponds to a view where the viewpoint and interest point are parallel to the X-axis on the XY-plane, with the viewpoint on the positive side. Increasing values for the azimuth correspond to rotating the view counter-clockwise around the interest point. Elevation is the angle of the viewpoint relative to the interest point on the XZ-plane. An elevation of 0 corresponds to a view where the viewpoint and interest point are parallel to the X-axis on the XZ-plane, with the viewpoint on the positive side. Increasing values for the elevation correspond to rotating the view counter-clockwise around the interest point in the XZ-plane. Roll is the angle that describes how much the view is rotated to the right. A roll of 0 corresponds to a view for which up is positive on the Z-axis. 	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Specifies the azimuth of the viewpoint relative to the interest point, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

Param2	Specifies the elevation of the viewpoint relative to the interest point, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Specifies the roll of the view, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_AZIMUTH +	Sets the azimuth of the view.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the azimuth of the view relative to the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISTANCE +	Sets the distance between the viewpoint and the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the distance between the viewpoint and the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELEVATION +	Sets the elevation of the view relative to the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the elevation of the view relative to the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLIP	Sets the view to a position and orientation that is flipped around the interest point in the specified axis.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_X	Specifies to flip the viewpoint around the interest point in the specified axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_Y	Specifies to flip the viewpoint around the interest point in the specified axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_Z	Specifies to flip the viewpoint around the interest point in the specified axis.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTEREST_POINT +	Sets the position of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the X-coordinate of the position of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b

	Param2	Specifies the Y-coordinate of the position of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
	Param3	Specifies the Z-coordinate of the position of the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_ORBIT_HORIZONTAL +		Sets the angle of the viewpoint's horizontal orbit around the interest point. This angle is used relative to the M_UP_VECTOR of the view. For more information, see the Orbit, up-vector, and distance subsection of the Manipulating the view section of Chapter 43: 3D Display and graphics. Note that orbit is always specified relative to the current position of the view. For example, repeatedly using this function with M_ORBIT_HORIZONTAL and the value 90 will repeatedly orbit the viewpoint horizontally around the interest point, in 90 degree increments.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1	Specifies the angle of the viewpoint's horizontal orbit around the interest point.	c		e	f	g	a		i		j	k	l	m	n	b

M_ORBIT_VERTICAL +	Sets the angle of the viewpoint's vertical orbit around the interest point. This angle is used relative to the M_UP_VECTOR of the view. For more information, see the Orbit, up-vector, and distance subsection of the Manipulating the view section of Chapter 43: 3D Display and graphics.  Note that orbit is always specified relative to the current position of the view. For example, repeatedly using this function with M_ORBIT_VERTICAL and the value 90 will repeatedly orbit the viewpoint vertically around the interest point, in 90 degree increments.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the angle of the viewpoint's vertical orbit around the interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROLL +	Sets the roll of the view. This is an angle which defines what direction the top of the view will be pointing, with 0 degrees specifying that the view will be perfectly level. As this value increases, the view rolls to the right.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the roll of the view, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSLATE	Applies a translation to the viewpoint and interest point without altering their relative position and orientation.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the translation along the X-axis to apply to the positions of the viewpoint and interest point.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the translation along the Y-axis to apply to the positions of the viewpoint and interest point.	c		e	f	g	a		i		j	k	l	m	n	b

Param3	Specifies the translation along the Z-axis to apply to the positions of the viewpoint and interest point.	c		e	f	g	a		i		j	k	l	m	n	b
M_UP_VECTOR	Sets the vector which defines what direction the top of the view is pointing. Note that this setting is normalized and adjusted to a unit vector that is orthogonal to the current orientation of the view. If you use M3ddispGetView() with M_UP_VECTOR after setting the up vector, the returned value will therefore typically be different from the vector you specified.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the X-component of the up vector of the view.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the Y-component of the up vector of the view.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Specifies the Z-component of the up vector of the view.	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_BOX	Sets the viewpoint and interest point (without altering the orientation) such that the interest point is at the center of the box and the entire region defined by the box is within view. 	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_WHOLE_SCENE	Specifies to set the viewpoint and interest point so that everything in the scene is within the view.	c		e	f	g	a		i		j	k	l	m	n	b

3D box geometry object ID	Specifies the identifier of the 3D box geometry object used to set the 3D display's view. The 3D box geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY, and must have been successfully defined.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0	Specifies the scale factor. The position for the viewpoint is calculated as though the box is scaled by this factor.	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_MATRIX +	Sets the view to match a transformation matrix. If the transformation matrix is the identity matrix, the viewpoint will be at (0,0,1), with the interest point at the origin (0,0,0) and an up-vector of (0,1,0). This can also be thought of as a direct top-down view, with the up-vector aligned with the Y-axis. You must specify the Aurora Imaging Library identifier of a similarity transformation matrix. You can determine whether a transformation matrix is a similarity transformation matrix using M3dgeoInquire() with M_SIMILARITY.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the following.	c		e	f	g	a		i		j	k	l	m	n	b
MatrixId	Specifies the Aurora Imaging Library identifier of a similarity transformation matrix (previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX).	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEW_ORIENTATION +	Sets the direction in which the view is looking. This moves the viewpoint (or interest point if combined with M_MOVE_INTEREST_POINT) so that line of sight matches a given orientation, specified using either a predefined orientation or a vector that defines the direction from the viewpoint to the interest point.	c		e	f	g	a		i		j	k	l	m	n	b

	Param1	Specifies one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
		M_BOTTOM_TILTED	Specifies to set the orientation of the view to an interpolation between the bottom, front, and left views.	c		e	f	g	a		i		j	k	l	m	n	b
		M_BOTTOM_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly below the interest point (parallel to the Z-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_FRONT_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly in front of the interest point (parallel to the Y-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_LEFT_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly to the left of the interest point (parallel to the X-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_REAR_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly behind the interest point (parallel to the Y-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_RIGHT_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly to the right of the interest point (parallel to the X-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		M_TOP_TILTED	Specifies to set the orientation of the view to an interpolation between the top, front, and left views.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TOP_VIEW	Specifies to set the orientation of the view so that the viewpoint is directly above the interest point (parallel to the Z-axis of the working coordinate system of the 3D display).	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the X-component of the vector that defines the orientation of the view.	c		e	f	g	a		i		j	k	l	m	n	b

Param2	Specifies the Y-component of the vector that defines the orientation of the view.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Specifies the Z-component of the vector that defines the orientation of the view.	c		e	f	g	a		i		j	k	l	m	n	b
M_VIEWPOINT +	Sets the position of the view.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the X-coordinate of the position of the view.	c		e	f	g	a		i		j	k	l	m	n	b
Param2	Specifies the Y-coordinate of the position of the view.	c		e	f	g	a		i		j	k	l	m	n	b
Param3	Specifies the Z-coordinate of the position of the view.	c		e	f	g	a		i		j	k	l	m	n	b
M_ZOOM +	Zooms the view in or out by a factor. This is equivalent to dividing M_DISTANCE by the specified factor.	c		e	f	g	a		i		j	k	l	m	n	b
Param1	Specifies the factor by which to zoom the view. This must be set to a value greater than 0.0.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_AZIM_ELEV_ROLL; M_AZIMUTH; M_DISTANCE; M_ELEVATION; M_INTEREST_POINT; M_ROLL; M_VIEW_MATRIX; M_VIEWPOINT.

You can add the following value to the above-mentioned values to specify that the viewpoint is moved relative to the current position and orientation of the viewpoint..

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Specifies whether the motion is absolute or relative.																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_COMPOSE_WITH_CURRENT	Specifies that the view is altered relative to the current position and orientation of the view.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_AZIM_ELEV_ROLL; M_AZIMUTH; M_DISTANCE; M_ELEVATION; M_ORBIT_HORIZONTAL; M_ORBIT_VERTICAL; M_VIEW_ORIENTATION; M_ZOOM.

You can add the following value to the above-mentioned values to specify that the interest point is moved instead of the viewpoint..

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Specifies whether the interest point should move																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_MOVE_INTEREST_POINT	Specifies that the interest point is moved instead of the viewpoint.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3d11.lib.
DLL	Requires ail11.dll; ail3d11.dll.